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RESEARCH ARTICLE

MangoLeafXNet: An Explainable Deep Learning Model for Accurate Mango Leaf Disease Classification

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ABSTRACT Addressing the global challenge of ensuring a consistent and abundant supply of fresh fruit, particularly in the context of fruit crops, is hindered by the prevalence of plant diseases. These diseases directly impact the quality of fruits, leading to a decline in overall agricultural production. Mango leaf diseases pose significant threats to global mango production, necessitating accurate and efficient classification techniques for timely disease management. Our study focuses on introducing MangoLeafXNet, a customized Convolutional Neural Network (CNN) architecture specifically tailored for the classification of mango leaf diseases, along with a healthy class. Our proposed model comprises six layers optimized to capture intricate disease patterns, demonstrating superior performance compared with prevalent pre-trained models. The model is trained and evaluated on three publicly available datasets: MangoLeafBD (4000 images across 8 classes), MangoPest (16 pest classes including healthy leaves), and MLDID (3000 high-resolution images across 5 classes). Our model demonstrated exceptional classification performance, attaining 99.8% accuracy, 99.62% recall, 99.5% precision, and an F1-score of 99.56%. Further validation on the MangoPest dataset and the Mango Leaf Disease Identification Dataset (MLDID) resulted in accuracies of 96.31% and 96.33%, respectively, confirming the robustness and adaptability of MangoLeafXNet across different datasets. Additionally, we incorporate Explainable AI techniques, including GRAD-CAM, Saliency Map, and LIME to enhance the interpretability of our model. We deployed Gradio web interface to create an interactive interface that allows users to upload images of mango leaves and get real-time classification and validation results along with confidence scores. This contribution not only advances the state-of-the-art in mango leaf disease classification but also offers promising prospects for real-time disease diagnosis and precision agriculture applications, contributing to enhanced crop health monitoring and sustainable mango cultivation practices.

INDEX TERMS Image processing, convolutional neural network, deep learning, mango leaf disease, image classification, explainable AI.

I. INTRODUCTION

The fruit of *Mangifera indica* L., celebrated for its flavor and nutritional value, is vital in agriculture and international

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trade [1]. However, mango cultivation faces challenges from foliar diseases that cause severe yield losses and quality degradation. These diseases, caused by bacteria, fungi, and viruses, primarily affect leaves, reducing photosynthetic efficiency and overall plant health, making timely diagnosis crucial [2], [3]. The complexity of managing these diseases

requires robust diagnostic and management strategies to mitigate their detrimental effects, ensuring the health of mango orchards and maintaining the quality of their produce. The sustainability and financial success of the global mango industry depend heavily on efficient disease control.

Widely used techniques for diagnosing leaf diseases in mango plants predominantly depend on physical evaluation by agricultural experts. This approach, while time-honored, often results in subjective assessments that can vary significantly between different inspectors, potentially leading to inconsistent diagnoses and delayed interventions. These delays can exacerbate the spread of diseases, further compromising yield and fruit quality. The emergence of computer vision and machine learning (ML) techniques has revolutionized this field by introducing automated disease classification systems that promise rapid and objective diagnosis [4]. These advanced systems facilitate timely treatments and preventive measures, significantly improving disease management outcomes [5], [6]. Among the various ML approaches, Convolutional Neural Networks (CNNs) are highly efficient in handling image classification. CNNs utilize hierarchical feature extraction to identify and analyze subtle patterns within images, thereby enabling them to accurately differentiate between normal and infected leaf specimens. This ability to discern fine details makes CNNs particularly powerful in addressing the complexities of mango leaf disease classification [7]. By automating the diagnostic process, these technologies not only enhance accuracy but also expedite intervention, thereby supporting more sustainable and productive mango cultivation.

Despite significant progress in CNN-based disease classification, current models frequently encounter challenges related to scalability, efficiency, and generalizability [8]. Although pre-trained CNN architectures are highly effective for general image classification tasks, they often fall short in terms of specificity and robustness when faced with domain-specific challenges like mango leaf disease classification [9]. These models may not be finely tuned to detect the nuanced patterns and variations characteristic of specific plant diseases, leading to less accurate diagnoses. Additionally, the training of deep neural networks is computationally intensive and requires substantial processing power and memory [10], [11]. This high computational demand poses a considerable obstacle to practical application, especially in resource-constrained agricultural settings where access to advanced computing infrastructure is limited. Consequently, the deployment of these sophisticated algorithms applied in practical environments are often hindered, underscoring the need for more efficient and scalable solutions tailored to the unique requirements of agricultural disease management.

In response to these challenges, our study introduces MangoLeafXNet, an innovative CNN architecture designed not only for the classification of mango leaf diseases, including healthy leaves, but also adaptable for broader agricultural applications. This architecture is meticulously optimized to capture intricate disease patterns with high

precision and efficiency. By fine-tuning the network structure and training parameters, MangoLeafXNet achieves enhanced classification accuracy while significantly reducing computational complexity and training time. Additionally, our model is lightweight, making it ideal for deployment in resource-constrained agricultural settings. MangoLeafXNet is also versatile, capable of adapting to other agricultural datasets to improve classification accuracy across a variety of crop diseases. Furthermore, it integrates Explainable AI (XAI) methods, providing transparency in the decision-making process and allowing users to understand the rationale behind each classification. This approach not only advances current methodologies in leaf disease classification but also addresses the practical limitations of resource-intensive training, facilitating wider implementation in real-world agricultural environments.

Overall, this study's key contributions include:

- We propose MangoLeafXNet, a customized CNN architecture designed for analyzing and classifying diseases found in mango leaves.
- We evaluate the performance of our model across three diverse datasets: MangoLeafBD, MangoPest, and Mango Leaf Disease Identification Dataset (MLDID), demonstrating its effectiveness and generalizability in different agricultural contexts.
- We apply XAI approaches, specifically Gradient-weighted Class Activation Mapping (GRAD-CAM), Saliency Map and Local Interpretable Model-agnostic Explanations (LIME), to improve the model's interpretability and transparency in decision-making.
- We develop an interactive Gradio-based web interface that enables real-time mango leaf disease classification, allowing users to upload images and receive instant predictions.

The structure of this paper is as follows: Section II analyzes the related research articles and their research gaps. Then, Section III outlines the methodology we followed to propose the MangoLeafXNet model, and Section IV discusses the experimental results of our proposed model and compares it with other existing literature and pre-trained models. After that, Section V discusses the overall effectiveness of the model along with its limitations. Finally, Section VI provides the conclusion of the study.

II. RELATED WORK

Significant progress has been made in plant disease classification through the adoption of deep learning (DL) methods, particularly with the use of CNNs. Several studies have demonstrated the effectiveness of CNNs in accurately diagnosing plant diseases from leaf images. For example, in a pivotal study by Sladojevic et al., the researchers developed a sophisticated CNN-based model specifically designed for plant disease classification. This model demonstrated remarkable accuracy, significantly enhancing the ability to distinguish between various plant diseases through detailed

analysis of leaf images. Their work highlighted the potential of CNNs to capture complex patterns and features associated with different plant diseases, setting a new standard for automated plant disease diagnosis [12]. Similarly, the study conducted by Mohanty et al. further showcased the power of DL in the agricultural sector. By employing advanced deep neural networks, their research illustrated that these networks could exceed the performance of standard ML approaches in plant disease diagnosis. The DL models used in their study were able to learn and identify intricate disease features from large datasets of leaf images, leading to significantly improved diagnostic accuracy and robustness [13].

Furthermore, Prabu et al. proposed a novel framework employing a CNN model enhanced through a crossover-based Levy flight distribution technique for mango leaf disease classification, demonstrating superior classification performance across four classes, particularly with recall values up to 98.42% for Mango Anthracnose. However, limitations include potential computational complexity and the need for further enhancement using DL for increased speed and accuracy [14]. In another work, Rizvee et al. presented LeafNet, a custom CNN designed for the early detection of seven common mango leaf diseases in Bangladesh. Through extensive experimentation and evaluation, LeafNet achieves an impressive average accuracy of 98.55%, outperforming state-of-the-art models like VGG16 and AlexNet. Additionally, LeafNet boasts faster training times and lower computational complexity, making it suitable for deployment on low-resource devices such as smartphones. While LeafNet shows promising results, further research could explore fine-tuning hyperparameters and testing its performance on larger and more diverse datasets to validate its effectiveness across different regions and conditions [15].

Several pre-trained models have been extensively utilized in plant disease classification due to their robustness and high accuracy. Models such as VGGNet, ResNet, AlexNet, and InceptionV3 stand out for their exceptional performance in various image classification tasks, including plant pathology. Ferentinos explored the application of DL models for detecting plant diseases across a broad spectrum of crops. Their research confirmed the effectiveness of these advanced architectures, such as VGGNet and ResNet, in accurately diagnosing plant diseases. The study highlighted the versatility and reliability of DL models, showcasing their ability to generalize across different types of plant diseases and agricultural applications [16]. Varma et al. investigated transfer learning with seven DL models to detect mango leaf diseases. Among them, InceptionV3 demonstrated the highest accuracy of 99.87%, outperforming other models. By leveraging pre-trained networks and extensive data augmentation techniques, the proposed approach shows promise for early and accurate detection of mango leaf diseases, potentially enhancing crop management and sustaining mango production [17]. Rao et al. presented an automated approach to classify mango leaf diseases using

DL and transfer learning techniques. The proposed system achieves accuracy rates of 89% for mango leaves, utilizing the pre-trained AlexNet model. The study demonstrates the feasibility of deploying a low-cost disease detection system on smartphones, potentially mitigating economic losses for farmers. However, limitations such as dataset diversity, early detection challenges, and real-time environmental factors underscore the need for further research and development in this domain [18]. These studies underscore the impact of pre-trained models in advancing plant disease classification, which has paved the way for more efficient and reliable agricultural practices, enabling better disease management and improving crop health and productivity.

Despite the success of these pre-trained models, they often come with high computational costs and complexities, which may not be ideal for real-time or resource-constrained environments. To address these challenges, several researchers have proposed custom CNN architectures tailored to specific tasks. For example, Too et al. introduced a lightweight CNN architecture designed for cassava leaf disease classification, which provided a balance between accuracy and computational efficiency [19]. In another work, Hari et al. presented a lightweight CNN model for detecting diseases in mango, guava, and banana leaves, utilizing only 101,000 parameters while achieving a high accuracy of 99.14%. The model significantly outperforms 15 state-of-the-art pre-trained models in terms of both accuracy and complexity, demonstrating its efficacy for use in resource-constrained devices [20].

Ensemble learning methods have gained popularity in plant disease classification for improving model stability and generalization. Choquet fuzzy integrals, a mathematical tool that aggregates the outputs of multiple classifiers by considering their interactions, have been applied to enhance ensemble approaches by integrating multiple deep learning models. A recent study by Hang et al. introduced an ensemble model for identifying tomato leaf diseases using Choquet fuzzy integrals, achieving 99.80% accuracy by combining pre-trained models with fuzzy aggregation. This method leverages classifier interactions to improve classification performance, offering valuable insights for mango leaf disease detection [21].

DL models have been integrated with frameworks like MapReduce, a distributed computing framework that processes large datasets by dividing tasks into smaller sub-tasks and running them in parallel across multiple machines, to manage large-scale data and enhance classification performance. Matrouk et al. presented a MapReduce-based framework for sequential association rule mining combined with deep learning for user categorization in retail, achieving 0.88 precision and other performance improvements. This demonstrates how MapReduce can efficiently handle big data, which could be similarly applied to large agricultural datasets for plant disease classification, enhancing accuracy in disease detection [22].

TABLE 1. Comparative analysis of existing methodologies in plant disease classification across different domains. This table summarizes the models, datasets, contributions, and challenges of various deep learning-based approaches, highlighting strengths and weaknesses in plant disease detection from leaf images.

Ref.	Model	Dataset	Contribution	Challenges
Sladojevic <i>et al.</i> [12]	Deep CNN (CaffeNet)	Leaf image dataset	Applied a DL method for accurate plant disease detection from leaf images.	Limited coverage of rare plant diseases, reliance on visible symptoms excluding early stages, lack of real-time field-deployable solutions, and risks of overfitting due to small samples in rare classes.
Mohanty <i>et al.</i> [13]	AlexNet, GoogLeNet (Transfer Learning)	PlantVillage dataset	Applied DL for accurate plant disease detection from leaf images, enabling smartphone-based diagnosis.	Reliance on visible symptoms, lack of real-time solutions, and overfitting due to small sample sizes. Multi-modal data integration remains a key research gap.
Prabu <i>et al.</i> [14]	CNN optimized by CROLFDD + MobileNetV2 + SVM	Mango leaf dataset (India)	Introduced a CNN-based framework for mango leaf disease classification optimized with a novel algorithm.	Limited validation, unexplored security risks, lack of interpretability, and gaps in scalability and domain adaptation.
Rizvee <i>et al.</i> [15]	LeafNet (CNN)	MangoleafBD dataset	Introduced LeafNet, a lightweight CNN excelling in detecting seven mango leaf diseases.	Limited research and datasets in Bangladesh hinder effective mango leaf disease detection models.
Varma <i>et al.</i> [17]	Transfer learning models (InceptionV3 best)	PlantVillage	Identifies InceptionV3 as the best model for mango leaf disease detection.	Need for diverse, real-field, and geographically varied datasets to improve model accuracy in agriculture.
Ferentinos <i>et al.</i> [16]	VGG (highest accuracy), AlexNetOWTBn	Open database of 87,848 images.	Developed CNN-based DL models for plant disease classification.	Performance drops with new data sources, also limited testing in real conditions.
Naik <i>et al.</i> [18]	Transfer Learning (VGG16, ResNet50)	Custom grape and mango datasets	Contributes to detecting mango and grape leaf diseases using transfer learning for precision farming in DL.	Performance reduced on images with poor lighting or extensive overlapping leaves.
Too <i>et al.</i> [19]	DenseNet 121, ResNet 50/101/152	PlantVillage (54,306 images)	Analyzed fine-tuning state-of-the-art DL architectures for accurate plant disease classification.	High computational time for ResNet 152 and Inception V4.
Hari <i>et al.</i> [20]	CNN	Mixed (Kaggle, Mendeley) for banana, guava, mango	Introduced a lightweight CNN model for banana, guava, and mango leaf disease detection.	Limited to banana, guava, and mango; further validation needed for other leaves.
Hang <i>et al.</i> [21]	Fuzzy Ensemble	PlantVillage	Introduced an ensemble model for identifying tomato leaf diseases using Choquet fuzzy integrals, integrating multiple pre-trained models with fuzzy aggregation.	The aggregation method lacks flexibility and richness, with potential flaws such as not considering classifier input correlations. Also, sensitivity to outliers.
Matrouk <i>et al.</i> [22]	TWDO-SAM + DCNN	Frequent Itemset Mining Dataset	Proposed a MapReduce-based framework for combining sequential association rule mining with deep learning for user categorization.	Handling large-scale datasets and efficiently processing them while maintaining model accuracy in real-time applications remains a challenge.

Additionally, self-supervised learning (SSL) methods have been explored in other domains to address challenges such as label sparsity and enhance model generalization. These approaches have shown promise in improving feature extraction and performance, particularly when labeled data is limited or difficult to obtain [23]. By leveraging SSL, models can learn more robust representations from unlabeled data, which could be beneficial in agricultural applications like mango leaf disease classification, where acquiring large

annotated datasets can be costly and time-consuming [24]. Table 1 summarizes the performance of various deep learning algorithms used in plant disease detection, outlining their contributions, strengths, and challenges.

III. METHODOLOGY

This section describes the methodology employed in this study, encompassing data collection methods, pre-processing techniques, the integration of MangoLeafXNet.

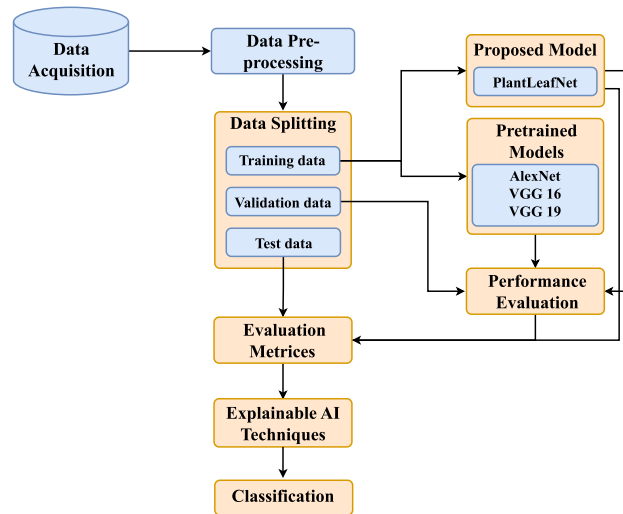


FIGURE 1. Workflow of the mango leaf disease classification process, including data acquisition, preprocessing, data splitting, and model evaluation using pretrained models and MangoLeafXNet.

A step-by-step representation of the proposed approach is provided in figure 1.

A. DATA ACQUISITION AND PRE-PROCESSING

1) MANGOLEAFBD

Our research primarily focuses on the MangoLeafBD dataset [25], the first comprehensive collection of mango leaf disease data in Bangladesh, sourced from four distinct orchards across the country: Sher-e-Bangla Agricultural University orchard (Dhaka), Jahangir Nagar University orchard (Savar), Udaypur village orchard (Rajbari), and Itakhola village orchard (Nilphamari). This dataset contains 4000 mango leaf images in JPEG format, obtained through mobile photography at these orchards, with each image having a resolution of 240×320 pixels. It is a well-balanced dataset, consisting of eight distinct classes, including healthy mango leaves and seven types of mango diseases: Sooty Mould, Gall Midge, Die Back, Cutting Weevil, Anthracnose, Powdery Mildew, and Bacterial Canker. Each class contains 500 image samples, ensuring an even distribution of data across all categories. An illustrative representation of the dataset, along with the list of corresponding class names, is provided in figure 2.

The study employs several crucial preprocessing steps to prepare the image dataset for training and validation. Firstly, the dataset is split into training, testing, and validation sets using a ratio of 80-10-10. To maintain consistency in input size, all images are uniformly resized to 227×227 pixels. Normalization is applied to standardize pixel values uniformly throughout the dataset. Additionally, data augmentation techniques such as horizontal flipping, random rotation, width and height shifting, zooming, and brightness_range are utilized. These techniques serve to enhance

dataset diversity, enabling the model to generalize effectively to various object orientations and scenarios. While the dataset is well-balanced, potential biases could arise from the dataset being sourced from only four orchards in specific regions of Bangladesh, which may not fully capture the variability of mango diseases across different environmental conditions or regions. To address potential image noise and enhance the quality of the dataset, we applied Gaussian Blur and CLAHE (Contrast Limited Adaptive Histogram Equalization). Gaussian Blur helps reduce small artifacts or noise in the images, while CLAHE improves contrast and makes disease features more distinguishable, thereby improving the model's ability to extract meaningful features and generalize better.

2) MANGO PEST CLASSIFICATION

For further evaluation, we utilized the dataset for Pest Classification in Mango Farms from Indonesia [26]. This dataset consists of images captured from mango farms affected by 15 different pests, as well as one category representing healthy, uninfected mango leaves. These pests can be identified through the structural and visual deformities they cause in the leaves. Images were acquired using low-cost sensors, which are widely used by farmers for on-site data collection. Most of the images have a resolution of 500×375 pixels. The dataset is imbalanced, with an unequal distribution of samples across the various classes. A visual representation of the dataset is shown in figure 3.

To mitigate class imbalance in the dataset, we employed the oversampling method. This method increases the sample count of classes with fewer samples to align with the class containing the highest number of samples. In our case, the class *mictis_longicornis* contains the highest number of

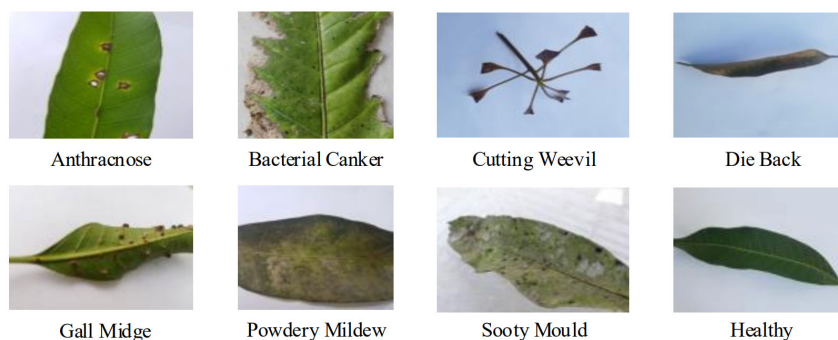


FIGURE 2. Overview of the MangoLeafBD dataset, illustrating the distribution of different mango leaf disease classes and healthy leaf samples.

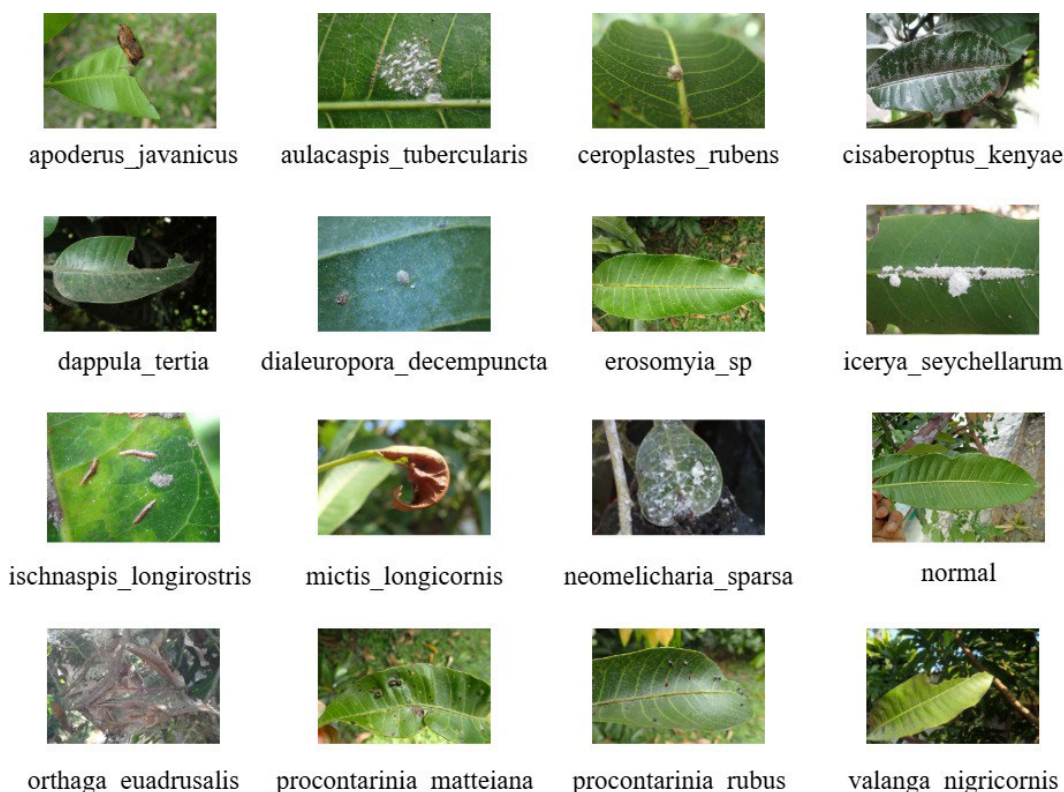


FIGURE 3. Overview of the Mango Pest dataset, showcasing the distribution of various pest categories affecting mango leaves, along with the normal class.

images (86). As a result, we oversampled the other classes to have 86 samples each, ensuring a balanced dataset. The visual representation of the dataset before balancing and after balancing is illustrated in figure 4. The number of samples per class before and after oversampling is provided in Table 2. We have also resized the images to 227×227 pixels to ensure consistency in image dimensions across the dataset, which is a standard input size for many DL models. Normalization is applied to standardize the pixel values, ensuring that all images have consistent intensity values, which helps improve the model's performance and convergence. Additionally, data augmentation methods such as horizontal flipping, random rotation, width and height shifting, and zooming are utilized

to artificially increase the variability of the dataset. These techniques mitigate overfitting by introducing more diverse training examples, enhancing the model's robustness to variations in image orientation and scale. To further enhance model training and evaluation, we split the dataset into 80% for training, 10% for testing, and 10% for validation, ensuring that the model has sufficient data for both training and performance evaluation.

3) MANGO LEAF DISEASE IDENTIFICATION DATASET (MLDID)

To assess the performance of the model on a third dataset, we used the Mango Leaf Disease Identification Dataset

TABLE 2. Class distribution before and after oversampling, showing the adjustment of class sizes to ensure balanced data.

Class Name	Data Before Oversampling	Data After Oversampling
ischnaspis_longirostris	07	86
ceroplastes_rubens	11	86
aulacaspis_tubercularis	13	86
erosomyia_sp	13	86
cisaberoptus_kenyae	19	86
orthaga_euadrusalis	22	86
procontarinia_rubus	26	86
icerya_seychellarum	27	86
procontarinia_matteiana	27	86
dialeuropora_decempuncta	30	86
neomelicharia_sparsa	36	86
normal	36	86
dappula_tertia	38	86
valanga_nigricornis	58	86
apoderus_javanicus	63	86
mictis_longicornis	86	86

(MLDID) [27]. This dataset is a comprehensive collection of high-quality images of mango leaves, carefully gathered from various regions of Bangladesh. It consists of 5 classes: Gall Midge, Healthy Leaves, Die Back, Bacterial Canker, and Anthracnose. Each class contains 600 images, totaling 3000 images. The images have a resolution of 512×512 pixels, ensuring high clarity and detail, which is crucial for effective analysis in various technological applications such as disease detection and classification. To better understand the class distribution and gain insights into the dataset, we also visualized the number of samples per class, which is shown in figure 5.

The preprocessing techniques used for this dataset are almost identical to those applied in the first two datasets. The dataset was divided into 80% for training, 10% for validation, and 10% for testing to ensure sufficient data for both training and evaluation. To enhance the model's generalization capability and prevent overfitting, data augmentation methods were applied during training, including random rotations, zooming, width shifts, horizontal flipping, and height shifts. These augmentations introduce variability in the dataset by simulating different orientations and scales. Additionally, normalization was applied to ensure consistent pixel value scaling, and all images were resized to 227×227 pixels to meet the model's input requirements.

B. PROPOSED MANGOLEAFXNET ARCHITECTURE

The proposed MangoLeafXNet is a deep CNN architecture designed for classifying mango leaf diseases. It is designed to be efficient and scalable while extracting complex disease-specific features. The architecture consists of six convolutional layers, which are separated by pooling layers, activation functions, and batch normalization, followed by fully connected layers for classification. The detailed layer-wise structure of the MangoLeafXNet model is presented in

Table 3, while figure 6 illustrates a visual representation of the overall architecture.

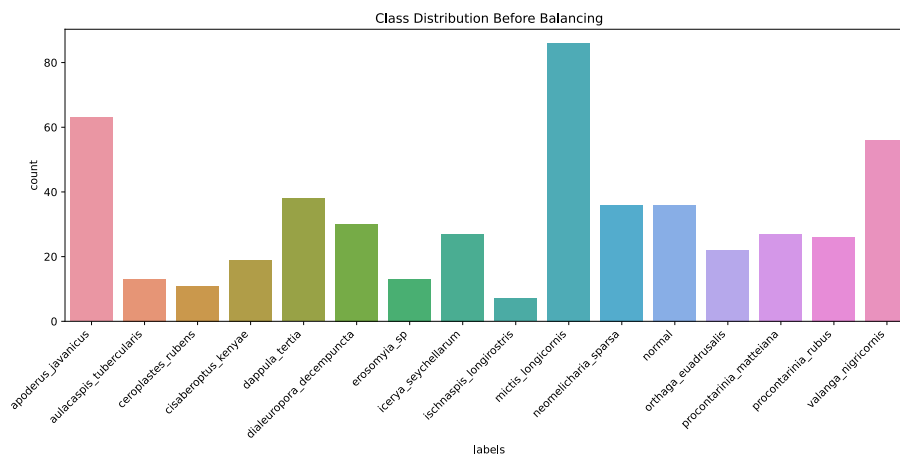
CONVOLUTIONAL AND FEATURE EXTRACTION LAYERS

The model starts with a convolutional layer of 7×7 with 64 filters and a 2 strides, capturing low-level image attributes like textures and edges. A batch normalization layer ensures stable activations, with a ReLU activation applied subsequently to incorporate non-linearity. Following layers employ 5×5 and 3×3 kernels, allowing the model to incrementally extract more complex patterns from the input images. Specifically:

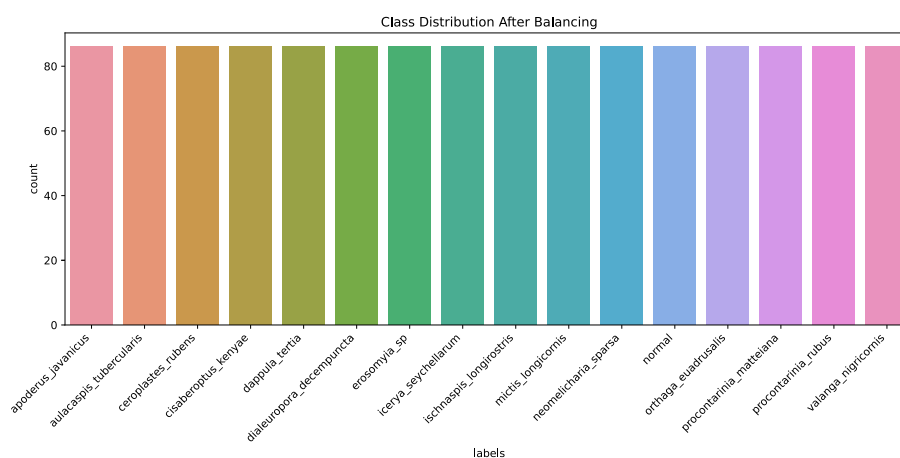
- **Second convolutional layer (128 filters, 5×5 kernel):** Refines feature extraction, followed by a max-pooling layer (3×3 , stride 2) to reduce spatial dimensions while preserving important features.
- **Third convolutional layer (256 filters, 3×3 kernel):** Enhances mid-level features such as disease-specific textures and leaf abnormalities.
- **Fourth and fifth layers (512 filters, 3×3 kernel each):** Further capture fine-grained disease patterns, helping the model distinguish between closely related disease classes.
- **Final 256-filter convolutional layer (3×3 kernel):** combines representations of deep features before sending them to the classifier.

CLASSIFICATION LAYERS

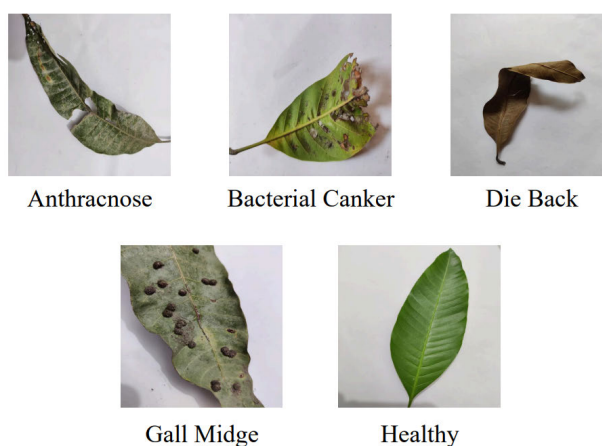
Extracted features are converted into a one-dimensional format before being fed into fully connected (Dense) layers with 512 neurons. Batch normalization and ReLU activation further enhance training stability. To mitigate overfitting, a dropout mechanism with a rate of 0.3 is incorporated. Finally, a softmax output layer with 8 neurons classifies the input image into one of 8 mango pest and disease categories.



(a) Dataset Before Balancing.



(b) Dataset After Balancing.

FIGURE 4. Visual representation of the dataset before and after balancing.**FIGURE 5.** Overview of the Mango Leaf Disease Identification (MLDID) dataset, illustrating the distribution of leaf disease classes, including Gall Midge, Healthy Leaves, Die Back, Bacterial Canker, and Anthracnose.

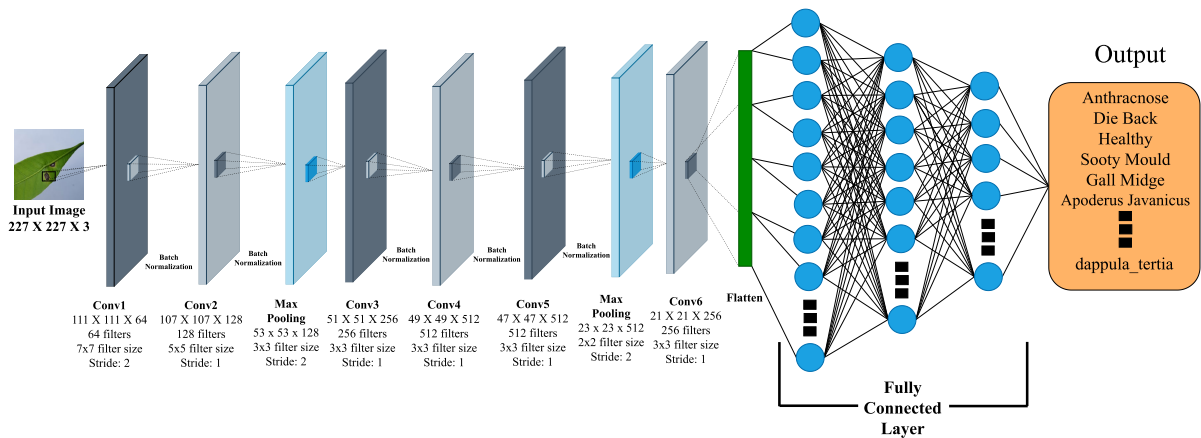
C. PRETRAINED MODELS FOR COMPARATIVE ANALYSIS

To evaluate the performance of the proposed MangoLeafXNet architecture, we compared it against several

widely recognized pretrained CNNs, namely AlexNet, VGG16, VGG19, ResNet50, and EfficientNetB0. These models were selected based on their architectural diversity,

TABLE 3. Feature maps used in the proposed MangoLeafXNet architecture, detailing the layers, filter sizes, strides, feature map sizes, and activation functions.

Layers	#Filters	Filter Size	Stride	Feature Map Size	Activation Function
Input	-	-	-	227x227x3	-
Conv1	64	7x7	2	111x111x64	ReLU
Batch	-	-	-	111x111x64	-
Normalization	-	-	-	-	-
Conv2	128	5x5	1	107x107x128	ReLU
Batch	-	-	-	107x107x128	-
Normalization	-	-	-	-	-
Max Pooling	-	3x3	2	53x53x128	-
Conv3	256	3x3	1	51x51x256	ReLU
Batch	-	-	-	51x51x256	-
Normalization	-	-	-	-	-
Conv4	512	3x3	1	49x49x512	ReLU
Batch	-	-	-	49x49x512	-
Normalization	-	-	-	-	-
Conv5	512	3x3	1	47x47x512	ReLU
Batch	-	-	-	47x47x512	-
Normalization	-	-	-	-	-
Max Pooling	-	2x2	2	23x23x512	-
Conv6	256	3x3	1	21x21x256	ReLU
Batch	-	-	-	21x21x256	-
Normalization	-	-	-	-	-
Flatten	-	-	-	112896	-
FC1	512	-	-	512	ReLU
Dropout	-	-	-	-	Rate = 0.3
FC2	512	-	-	512	ReLU
FC3	8	-	-	8	Softmax

**FIGURE 6.** Architecture of the proposed MangoLeafXNet model, showing the convolutional, pooling, normalization, and fully connected layers used for mango leaf disease classification.

historical significance, and proven effectiveness in image classification tasks. This comparison allows us to highlight the strengths of our model in terms of performance, efficiency, and scalability.

1) ALEXNET

AlexNet consists of eight layers with learnable parameters: five convolutional layers followed by three fully connected layers. It introduced key innovations such as ReLU activation, local response normalization (LRN), overlapping max pooling, and dropout regularization. The architecture follows:

- Conv1 → MaxPooling
- Conv2 → MaxPooling
- Conv3 → Conv4 → Conv5 → MaxPooling

- Fully Connected Layers (FC1 → FC2 → FC3)

AlexNet's use of larger convolutional filters in the early layers and a straightforward sequential architecture aligns closely with the initial design of our proposed model. Comparing MangoLeafXNet with AlexNet helps evaluate how a more modern and deeper variant of a similar structural philosophy performs in the context of plant disease classification.

2) VGG16

VGG16 contains 13 convolutional layers and 3 fully connected layers. It exclusively uses 3×3 convolution filters and 2×2 max pooling. The structure is arranged in blocks:

- Block 1: Conv $\times 2$ (64 filters) → MaxPooling

- Block 2: Conv $\times 2$ (128 filters) $\rightarrow$ MaxPooling
- Block 3: Conv $\times 3$ (256 filters) $\rightarrow$ MaxPooling
- Block 4: Conv $\times 3$ (512 filters) $\rightarrow$ MaxPooling
- Block 5: Conv $\times 3$ (512 filters) $\rightarrow$ MaxPooling
- Fully Connected Layers (FC1 $\rightarrow$ FC2 $\rightarrow$ FC3)

VGG16 employs a deep, uniform architecture with small 3×3 filters, similar to the later layers in our MangoLeafXNet model. This makes it an ideal benchmark for evaluating whether a more compact model like ours can achieve competitive performance with significantly fewer layers and parameters.

3) VGG19

VGG19 is an extension of VGG16, increasing depth with 16 convolutional layers. The filter sizes and structure remain the same, with an emphasis on deeper feature extraction.

- Similar to VGG16 but:
- Block 3, 4, and 5: Conv $\times 4$ instead of Conv $\times 3$

VGG19 expands on VGG16 by increasing network depth, allowing deeper feature extraction. Since MangoLeafXNet incorporates a moderately deep stack of convolutional layers with similar design elements, comparing it to VGG19 allows us to analyze the trade-offs between depth, complexity, and performance.

4) ResNet50

ResNet50 is a 50-layer deep network that introduces residual connections. These shortcuts enable training of deeper models by allowing gradients to pass through identity mappings.

- Initial Conv $\rightarrow$ MaxPooling
- Conv2_x block (3 residual units)
- Conv3_x block (4 residual units)
- Conv4_x block (6 residual units)
- Conv5_x block (3 residual units)
- Global Average Pooling $\rightarrow$ Dense Layer

ResNet50 provides a strong benchmark for deep architectures. Its use of residual learning represents a key innovation in CNN design, and it helps us compare against highly scalable models.

5) EfficientNetB0

EfficientNetB0 uses compound scaling to optimize depth, width, and resolution. It incorporates Mobile Inverted Bottleneck (MBConv) blocks and Squeeze-and-Excitation (SE) modules.

- Initial Conv $\rightarrow$ MBConv blocks (7 stages)
- Conv1 $\times 1 \rightarrow$ Global Average Pooling
- Dropout $\rightarrow$ Dense Layer (Softmax)

EfficientNetB0 is lightweight and achieves competitive accuracy with fewer parameters. Its efficient design allows comparison with MangoLeafXNet in resource-constrained settings.

TABLE 4. Training parameters used for the MangoLeafXNet model.

Name of the Parameter	Values
Batch Size	12 & 16
Epochs	50 & 100
Loss Function	Categorical Cross Entropy
Learning Rate	0.001 & 0.0005
Optimizer	SGD
Dropout Rate	0.3 & 0.4

D. EXPERIMENTAL SETUP

The MangoLeafXNet was employed to classify the mango leaf disease classes and healthy leaves. The proposed model was trained for 100 epochs on the MangoLeafBD and Mango Leaf Disease Identification Dataset (MLDID) with a batch size of 12 for MangoLeafBD and MangoPest, and a batch size of 16 for MLDID. The MangoPest dataset was trained for 50 epochs. Table 4 summarizes all the parameters utilized in this study. During training, the Stochastic Gradient Descent (SGD) optimizer was used with a learning rate of 0.001 for the MangoLeafBD and MangoPest datasets, and a learning rate of 0.0005 for the MLDID dataset, to minimize the loss and optimize the model. To evaluate performance comparatively, pretrained models including AlexNet, VGG16, VGG19, ResNet50, and EfficientNetB0 were trained using the same hyperparameter settings: batch size of 32, learning rate of 0.001, SGD optimizer, and 100 epochs. This consistent experimental setup ensures a fair evaluation, highlighting the performance differences clearly.

E. HYPERPARAMETER SELECTION

The selection of hyperparameters for MangoLeafXNet was guided by a combination of scientific experimentation and prior literature on CNN-based plant classification models. The batch size, dropout rate, and learning rate were fine-tuned through iterative experimentation to achieve an optimal trade-off between model generalization and training efficiency.

- **Batch Size:** Batch sizes of 12 and 16 were employed based on dataset size and memory constraints. A batch size of 12 was used for MangoLeafBD and MangoPest, while 16 was applied for MLDID. Smaller batch sizes (e.g., 8) resulted in higher variance in validation accuracy, while larger batch sizes (e.g., 32) slowed convergence without significant performance gains.
- **Dropout Rate:** Dropout was applied at two levels (0.3 and 0.4) to prevent overfitting and enhance generalization. Extensive testing across values between 0.2 and 0.5 revealed that 0.3 offered a stable regularization effect, while 0.4 provided additional robustness in deeper layers, ensuring improved model performance.
- **Learning Rate:** A learning rate of 0.001 was set for MangoLeafBD and MangoPest, whereas 0.0005 was applied for MLDID. These values were determined

through scientific evaluation, where higher learning rates (0.01) led to unstable convergence, and lower values (0.0001) resulted in excessively slow training. The chosen learning rates ensured steady optimization while preventing gradient divergence.

- **Hyperparameter Optimization Strategy:** Hyperparameters were optimized through manual iterative tuning rather than automated techniques like grid search or Bayesian optimization. Given the model size, manual tuning allowed for efficient testing across multiple configurations without excessive computational overhead.

IV. RESULT ANALYSIS

This section presents the findings from the experiments of our proposed MangoLeafXNet model and compares it with other existing literature and pre-trained models. The proposed MangoLeafXNet architecture was trained independently on each of the three datasets: MangoLeafBD, MangoPest, and MLDID. The same CNN configuration was maintained across all datasets without architectural modifications, with only minor adjustments to hyperparameters such as batch size and learning rate to match the specific dataset characteristics. This strategy highlights the model's flexibility and generalization ability across different types of mango leaf datasets.

A. QUANTITATIVE ASSESSMENT

Figures 7a - 7f present a side-by-side comparison of the training and validation accuracy across epochs for AlexNet (Figure 7a), VGG16 (Figure 7b), VGG19 (Figure 7c), ResNet50 (Figure 7d), EfficientNetB0 (Figure 7e), and our proposed model, MangoLeafXNet (Figure 7f). These figures illustrate how each model's accuracy evolves during the training process, highlighting their respective learning curves and generalization abilities. While AlexNet, VGG16, and VGG19 exhibit relatively stable and smooth convergence patterns, their learning curves indicate a gradual improvement in accuracy with fewer fluctuations between the training and validation phases. In contrast, ResNet50 and EfficientNetB0 exhibit noticeable fluctuations, particularly in their validation accuracy curves, suggesting susceptibility to overfitting and instability in certain epochs. MangoLeafXNet, as shown in figure 7f, is particularly noteworthy for its rapid convergence and consistent performance. Unlike the other models, MangoLeafXNet exhibits a stable accuracy with minimal fluctuations between the training and validation phases. This demonstrates that the model is efficiently adapting to the training data and generalizing well to the validation set, maintaining a robust performance throughout the training epochs. The consistency in the MangoLeafXNet's accuracy curve demonstrates its efficiency in capturing relevant features and its resilience to overfitting, ensuring reliable predictions on unseen data.

Figure 8 shows the performance of our proposed model, MangoLeafXNet, on the MangoPest Dataset as part of its validation alongside the MangoLeafBD Dataset. This assessment was undertaken to evaluate the model's effec-

tiveness and generalizability across diverse datasets. The consistent accuracy achieved on both datasets underscores MangoLeafXNet's robustness and adaptability, reinforcing its suitability for diverse agricultural applications and pest management scenarios.

After demonstrating strong performance on the MangoPest Dataset, Figure 9 illustrates the evaluation of MangoLeafXNet on the Mango Leaf Disease Identification Dataset (MLDID) to further validate its adaptability and effectiveness. The graph depicts the validation and training accuracy trends over 100 epochs. Despite some fluctuations in validation accuracy during the initial phases, the model maintains stable and high accuracy across both training and validation phases. This performance highlights MangoLeafXNet's robustness and reliability in accurately identifying mango leaf diseases, further confirming its applicability across diverse agricultural datasets and disease management tasks.

Figures 10a - 10f illustrate the training and validation loss across epochs for AlexNet (Figure 10a), VGG16 (Figure 10b), VGG19 (Figure 10c), ResNet50 (Figure 10d), EfficientNetB0 (Figure 10e), and the proposed model, MangoLeafXNet (Figure 10f). These figures provide insights into the optimization processes of each model, showing how well they minimize the loss during training and how effectively they generalize to validation data. It is evident that while AlexNet, VGG16, and VGG19 gradually reduce their loss with some degree of stability, they occasionally experience minor spikes, indicating fluctuations in learning. ResNet50 and EfficientNetB0 display more noticeable variations, especially in their validation loss, pointing to potential overfitting and instability during certain epochs. MangoLeafXNet, as depicted in figure 10f, is notable for its consistent reduction in both training and validation loss throughout the epochs. This steady decrease without significant divergence between the training and validation loss curves indicates that MangoLeafXNet maintains a balanced learning process, minimizing overfitting and ensuring robust performance. The visualization of the loss curves highlights the efficiency of the MangoLeafXNet's architecture in learning the complex patterns of mango leaf diseases, demonstrating its capability to maintain low loss values consistently.

Figure 11 presents the loss curves of training and validation of MangoLeafXNet on the MangoPest dataset, offering an understanding of the model's learning dynamics. Following the evaluation on the MangoLeafBD dataset, this additional test aimed to validate MangoLeafXNet's generalizability to pest-related data. As demonstrated in the graph, both training and validation loss steadily decrease over epochs, reflecting an efficient learning process and a well-generalized model. This behavior reinforces the model's robustness and adaptability, ensuring reliable performance in identifying mango pests across diverse agricultural environments.

After evaluating on two datasets, we have further evaluated on another dataset to test the generalizability of our model.

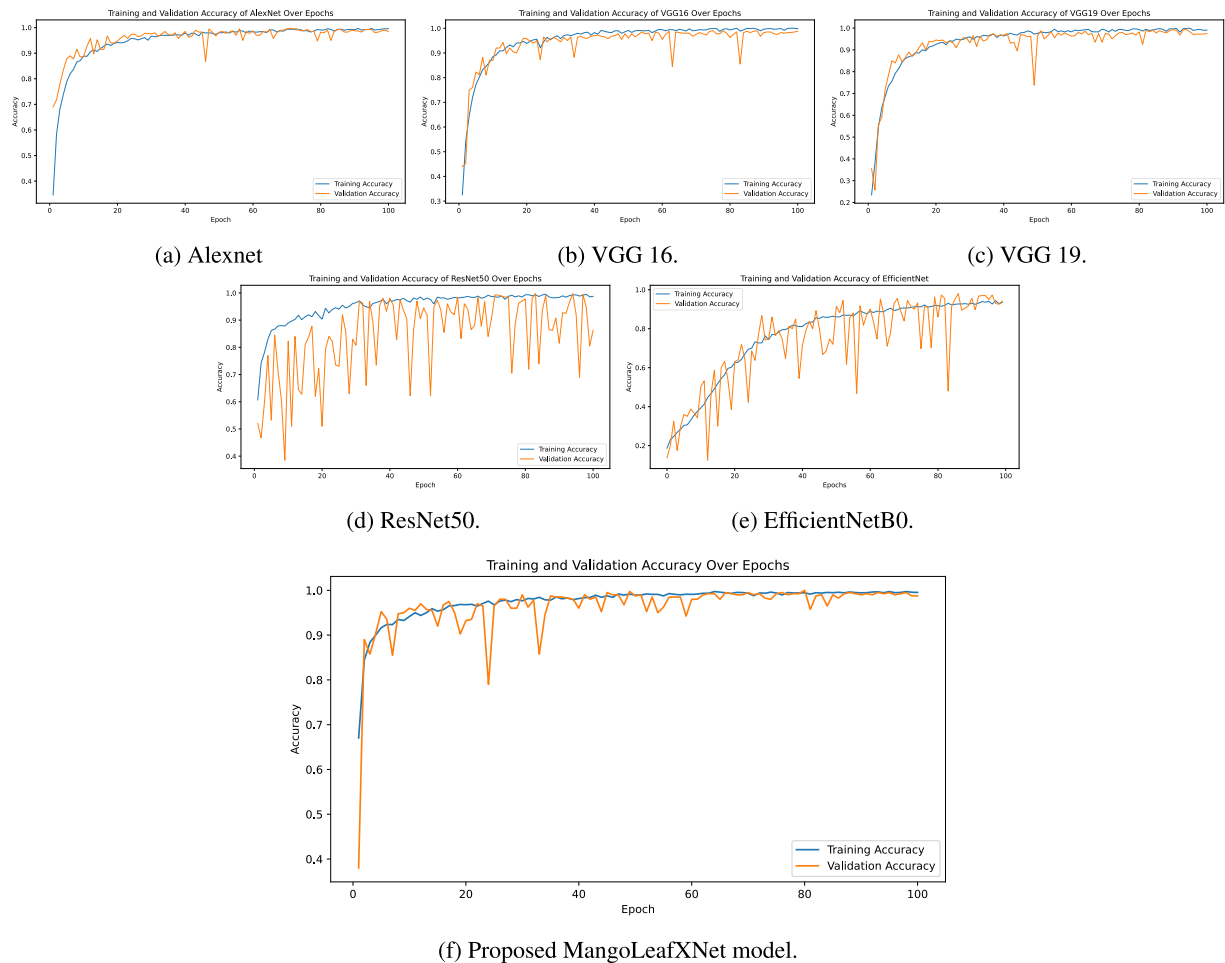


FIGURE 7. Performance comparison of training and validation accuracy across multiple epochs for different deep learning models using the MangoLeafBD dataset, including (a) AlexNet, (b) VGG16, (c) VGG19, (d) ResNet50, (e) EfficientNetB0, and (f) our proposed MangoLeafXNet model.

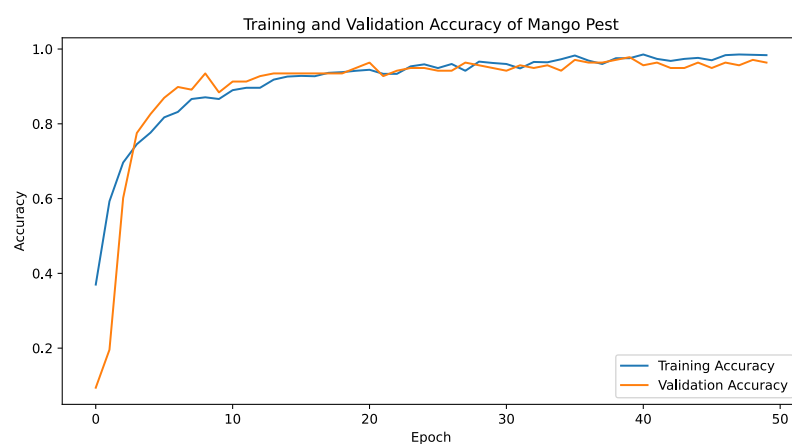


FIGURE 8. Training and validation accuracy for the proposed MangoLeafXNet model on the MangoPest Dataset.

Figure 12 shows the training and validation loss curves of MangoLeafXNet on the Mango Leaf Disease Identification Dataset (MLDID). The graph shows a steep drop in training

loss during the first few epochs, indicating fast learning by the model. However, the validation loss has a few ups and downs in the early stages, as the model adjusts to

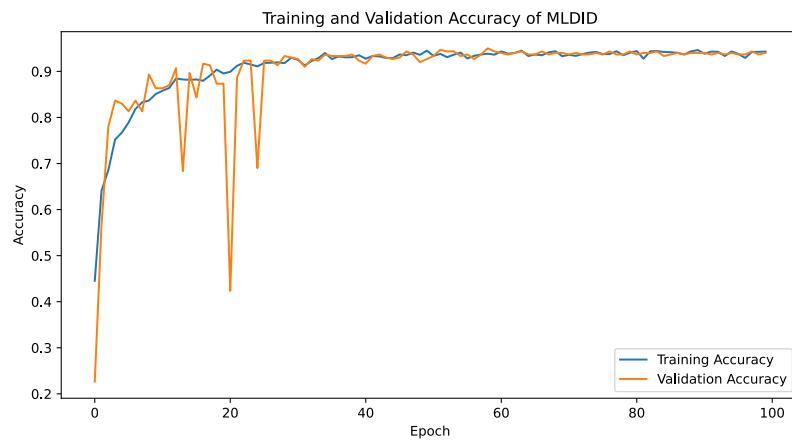


FIGURE 9. Training and validation accuracy for the proposed MangoLeafXNet model on the Mango Leaf Disease Identification Dataset (MLDID).

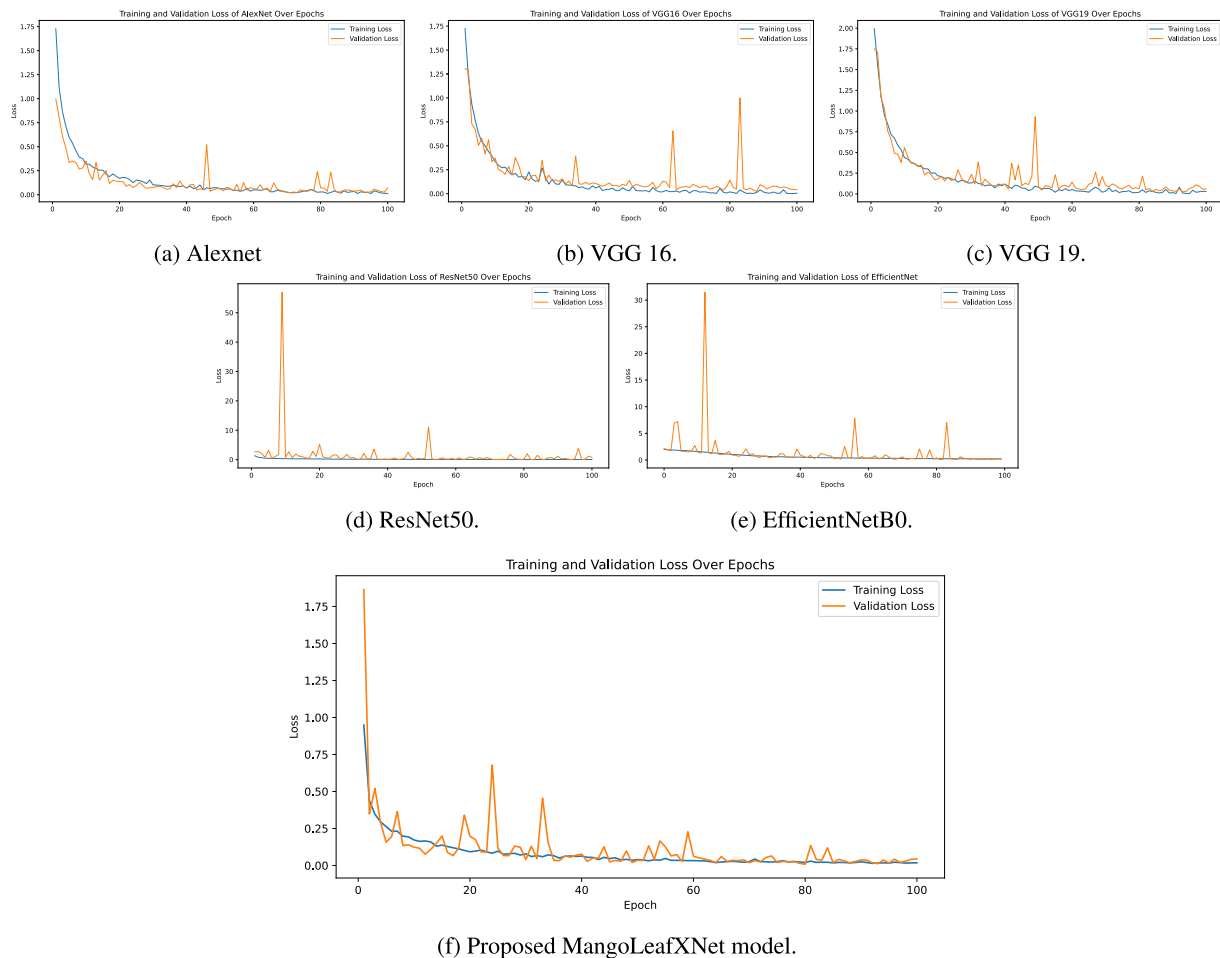


FIGURE 10. Performance comparison of training and validation loss across multiple epochs for different deep learning models using the MangoLeafBD dataset, including (a) AlexNet, (b) VGG16, (c) VGG19, (d) ResNet50, (e) EfficientNetB0, and (f) our proposed MangoLeafXNet model.

the dataset's complexity. Later in the training, both losses become stable and reach very low values, showing that the model has learned effectively without overfitting. This result confirms MangoLeafXNet's ability to perform well on

different datasets, proving its strength in identifying mango leaf diseases.

Figures 13a - 13f display the ROC curves for each class for AlexNet (Figure 13a), VGG16 (Figure 13b),

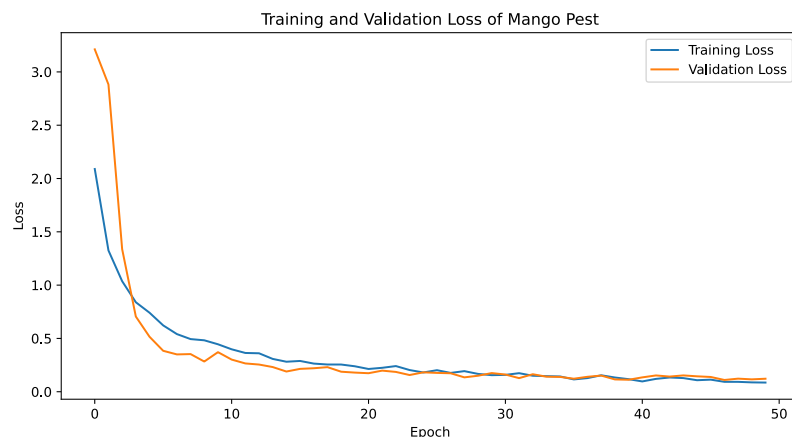


FIGURE 11. Training and validation Loss for the proposed MangoLeafXNet model on the MangoPest Dataset.

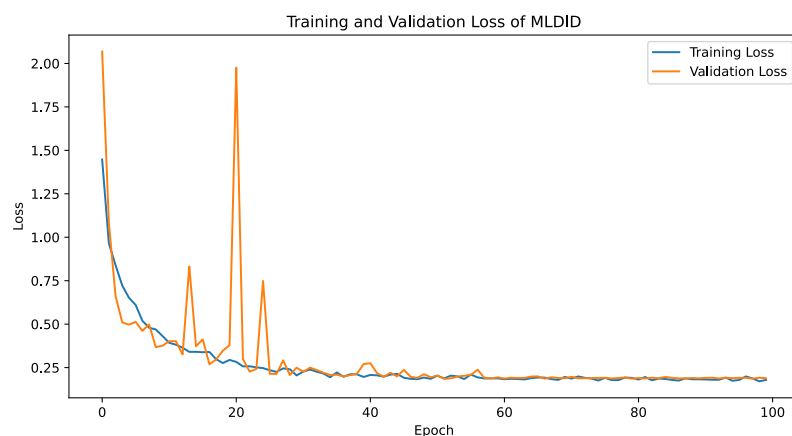


FIGURE 12. Training and validation Loss for the proposed MangoLeafXNet model on the Mango Leaf Disease Identification Dataset (MLDID).

VGG19 (Figure 13c), ResNet50 (Figure 13d), EfficientNetB0 (Figure 13e), and the proposed model, MangoLeafXNet (Figure 13f). These figures illustrate the true positive rate against the false positive rate for each disease class, providing a comprehensive view of each model's diagnostic performance. Among the models, MangoLeafXNet, shown in Figure 13f, stands out by achieving perfect AUC scores (1.00) across all classes, indicating its exceptional ability to differentiate between disease categories with both high sensitivity and specificity, making it particularly valuable for real-time disease identification in mango cultivation. In comparison, ResNet50 and EfficientNetB0 also exhibit strong ROC curves with near-perfect AUC values for most classes, reflecting their robust classification performance. However, minor deviations in certain classes for ResNet50 and EfficientNetB0 suggest slight inconsistencies in handling specific disease patterns, especially under challenging cases. This indicates that while these models perform well overall, they struggle with subtle or complex disease variations. Additionally, Figure 13 summarizes the ROC performance of

all the evaluated models on the MangoLeafBD dataset. While AlexNet, VGG16, and VGG19 demonstrate commendable results, ResNet50 and EfficientNetB0 achieve notably high AUC scores, reinforcing their effectiveness. Nonetheless, the proposed MangoLeafXNet outperforms all others by consistently reaching an AUC of 1.00 across all disease classes, highlighting its superior generalization and robustness in real-world disease identification scenarios. This perfect AUC across all disease classes confirms MangoLeafXNet's ability to reliably identify and distinguish between various mango leaf diseases.

Furthermore, figure 14 & 15 show the ROC curves and AUC scores for the proposed MangoLeafXNet model evaluated on the MangoPest and Mango Leaf Disease Identification Dataset (MLDID), respectively. In Figure 14, the ROC curves demonstrate exceptional performance, with AUC scores of 1.00 for most classes in the MangoPest dataset, and a slightly lower score of 0.99 for *Valanga nigricornis*. This highlights the model's robust classification ability across diverse pest categories. Although the slight dip

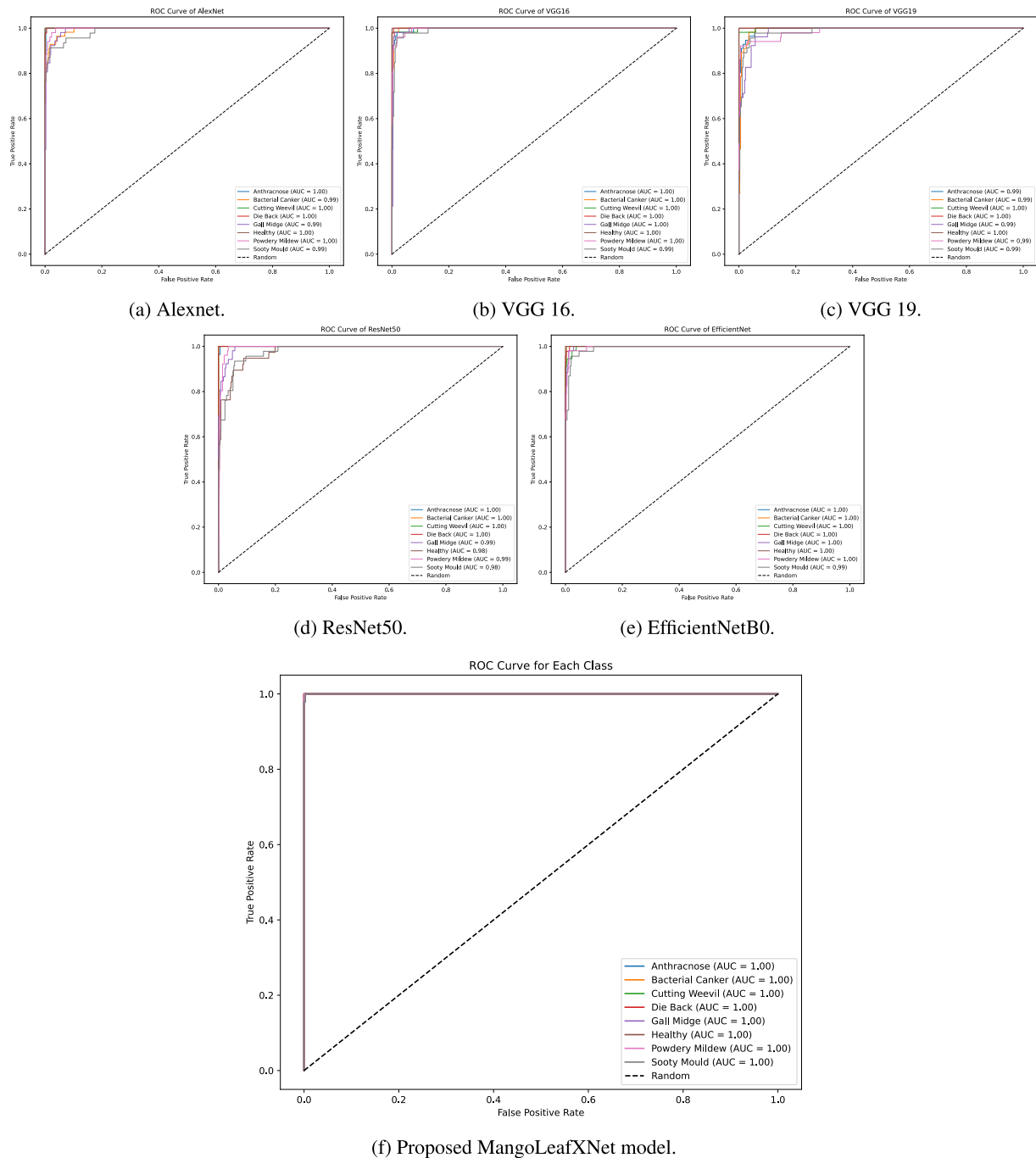


FIGURE 13. Comparison of AUC for different deep learning models using the MangoLeafBD dataset, including (a) AlexNet, (b) VGG16, (c) VGG19, (d) ResNet50, (e) EfficientNetB0, and (f) our proposed MangoLeafXNet model.

in AUC for a specific pest suggests room for improvement, it still confirms the model's overall robustness. Similarly, Figure 15 presents the ROC curves and corresponding AUC scores for the proposed MangoLeafXNet model on the Mango Leaf Disease Identification Dataset (MLDID). The model demonstrates strong classification performance, achieving high AUC values across all disease classes—such as 1.00 for Bacterial Canker, Die Back, and Healthy leaves, and slightly lower yet still robust scores for Anthracnose (0.98) and Gall Midge (0.97). The consistency in these results

highlights the model's capability to generalize effectively across different disease types and conditions, even with variations in the dataset. These findings further validate MangoLeafXNet's reliability and adaptability in real-world agricultural applications.

Figures 16a - 16f present the confusion matrices for AlexNet (Figure 16a), VGG16 (Figure 16b), VGG19 (Figure 16c), ResNet50 (Figure 16d), EfficientNetB0 (Figure 16e), and the proposed model, MangoLeafXNet (Figure 16f). These matrices provide detailed insights into the

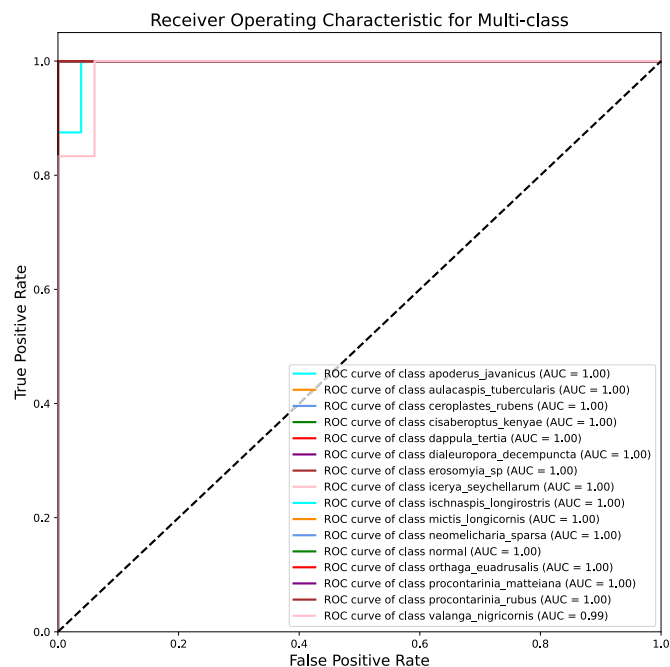


FIGURE 14. ROC curves and AUC scores for the proposed MangoLeafXNet model on the MangoPest dataset.

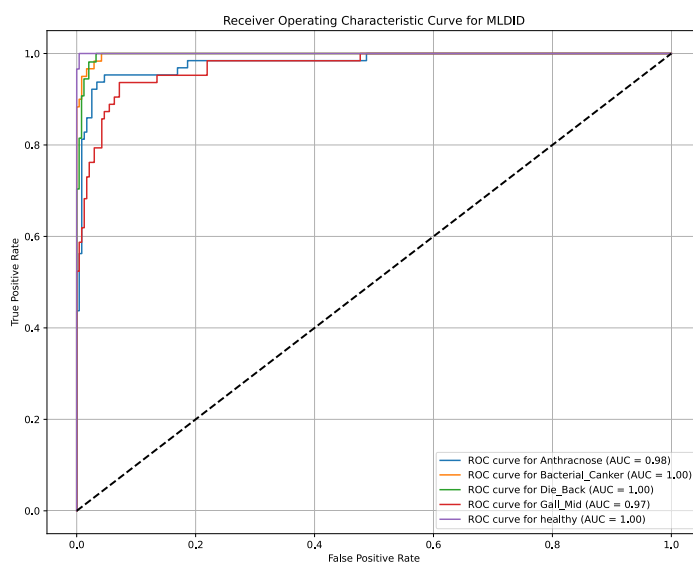


FIGURE 15. ROC curves and AUC scores for the proposed MangoLeafXNet model on the Mango Leaf Disease Identification Dataset (MLDID).

classification accuracy by displaying the true positives, true negatives, false positives, and false negatives for each class. Among the baseline models, AlexNet, VGG16, and VGG19 demonstrate relatively balanced performance with some minor misclassifications, particularly in differentiating between visually similar classes such as Sooty Mould and Bacterial Canker. ResNet50 and EfficientNetB0 show noticeable confusion, especially between Healthy leaves and classes like Gall Midge and Sooty Mould, indicating

potential overfitting or difficulty in generalizing across these categories. The MangoLeafXNet demonstrates exceptional performance with near-perfect classification across all classes. Specifically, for the Anthracnose class, the model correctly classified all samples, and similarly for Gall Midge, Healthy, Powdery Mildew, Cutting Weevil, Die Back, Bacterial Canker, all samples were accurately classified. However, there were minimal misclassifications observed in two classes: Sooty Mould, where 1 instance

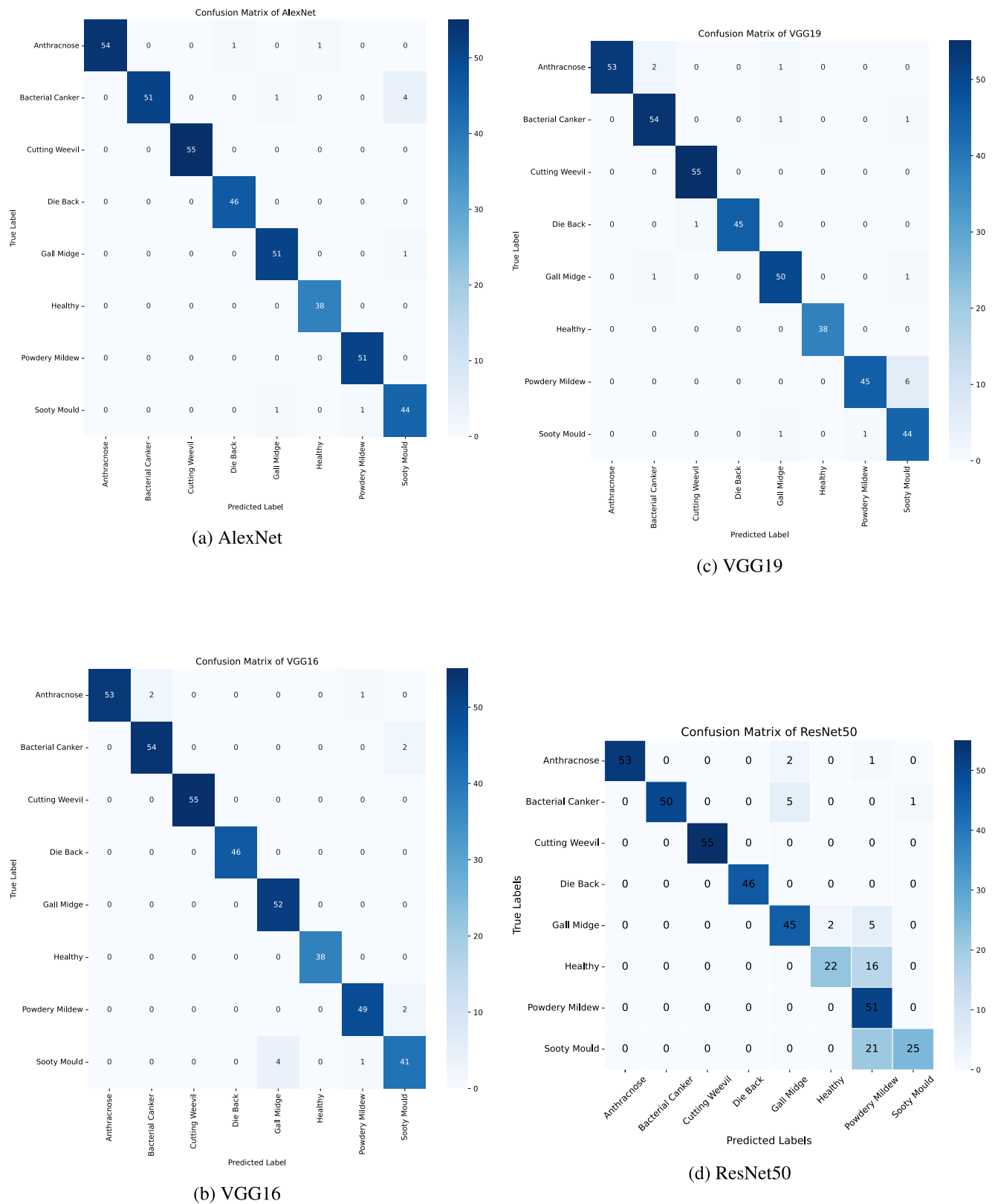


FIGURE 16. Comparison of confusion matrices for different deep learning models using the MangoLeafBD dataset: (a) AlexNet, (b) VGG16, (c) VGG19, (d) ResNet50, (e) EfficientNetB0, and (f) Proposed MangoLeafXNet model.

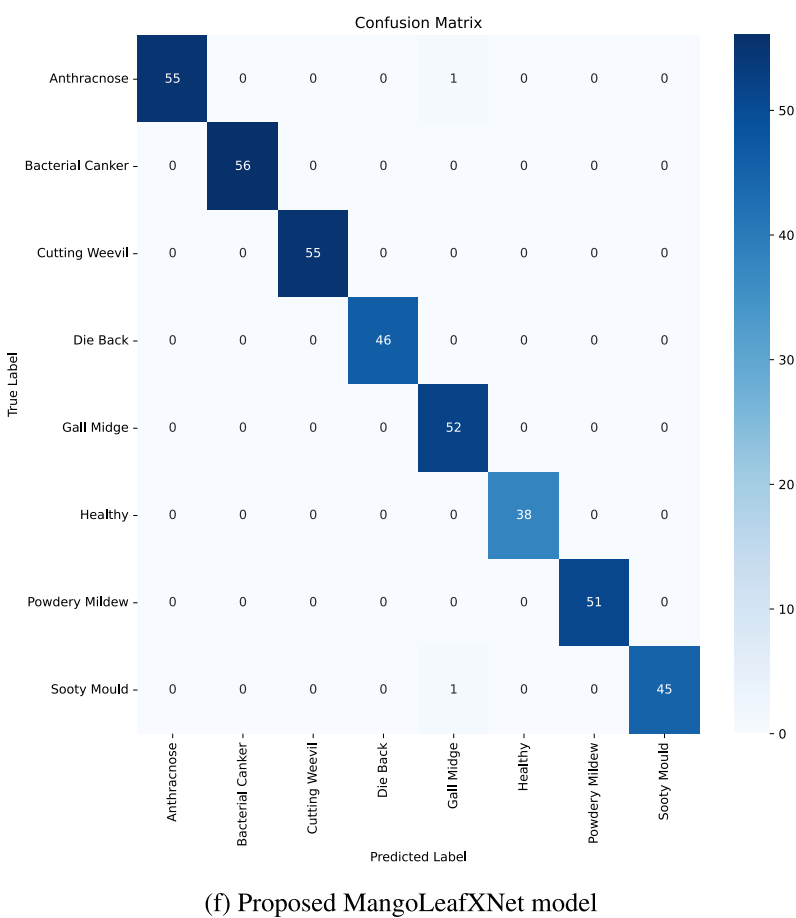
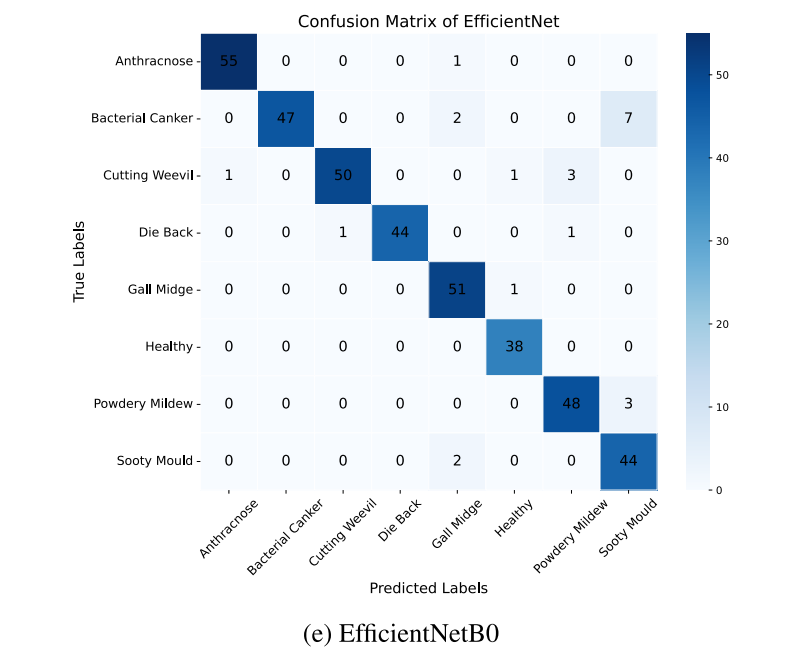


FIGURE 16. (Continued.) Comparison of confusion matrices for different deep learning models using the MangoLeafBD dataset: (a) AlexNet, (b) VGG16, (c) VGG19, (d) ResNet50, (e) EfficientNetB0, and (f) Proposed MangoLeafXNet model.

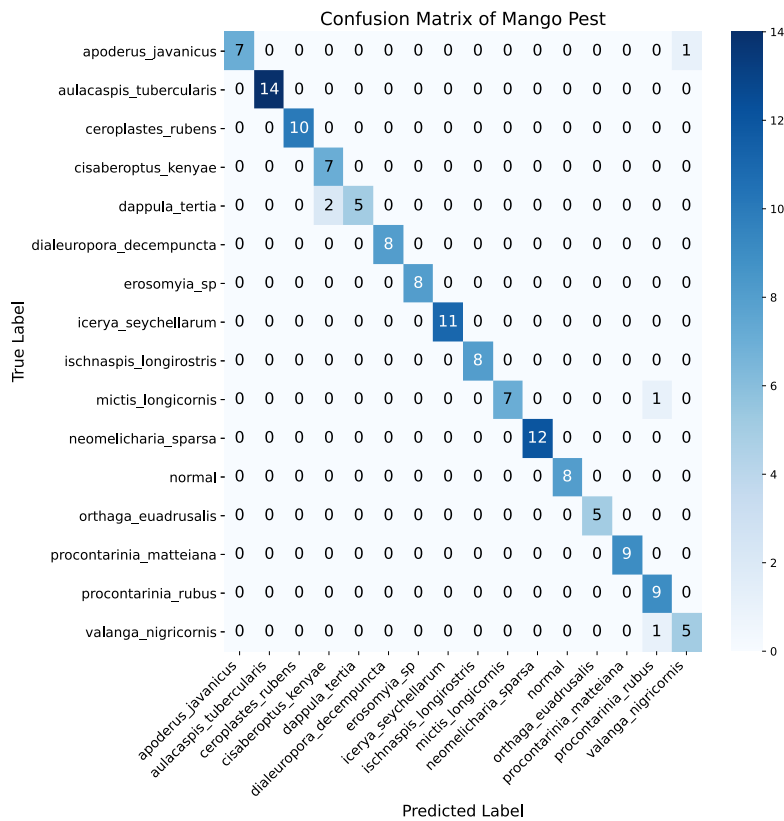


FIGURE 17. Confusion matrix for the proposed MangoLeafXNet model on the MangoPest dataset.

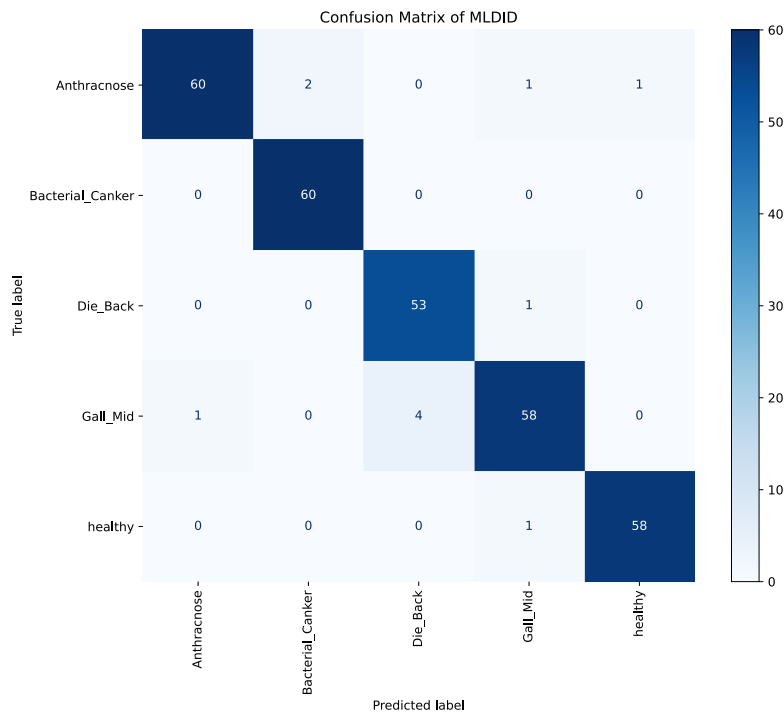


FIGURE 18. Confusion matrix for the proposed MangoLeafXNet model on the Mango Leaf Disease Identification Dataset (MLDID).

was incorrectly classified, and Healthy, where 1 instance was misclassified as Sooty Mould. Despite these minor

errors, MangoLeafXNet achieved correct classifications for all other classes, showcasing its superior accuracy and

reliability in distinguishing between different mango leaf diseases.

Figure 17 and Figure 18 shows the confusion matrices for the MangoLeafXNet model on the MangoPest and Mango Leaf Disease Identification Dataset (MLDID), respectively. These matrices illustrate the MangoLeafXNet's classification performance throughout various classes, showcasing its capacity to accurately differentiate between categories. The minimal misclassifications observed reinforce MangoLeafXNet's robustness and reliability in handling diverse datasets and distinguishing between different data types effectively.

Table 5 provides a performance comparison of various deep learning models—VGG16, AlexNet, VGG19, ResNet50, EfficientNetB0, and MangoLeafXNet—evaluated on the MangoLeafBD dataset for mango leaf disease classification. Among these, MangoLeafXNet demonstrated superior performance, achieving an overall evaluation score of 0.99, surpassing the other models. The model maintained consistent classification performance across various categories, attaining a perfect score of 1.00 across all key metrics for Die Back, Bacterial Canker, and Cutting Weevil. Among the various classified categories, 'Cutting Weevil' produced the most optimal outcome, obtaining a perfect 1.00 across all key performance metrics. On the other hand, while still demonstrating strong classification accuracy, 'Gall Midge' recorded a slightly lower performance with an F1-score of 0.98, influenced by its recall of 0.99 and precision of 0.96. Comparatively, ResNet50 and EfficientNetB0 exhibit fluctuating performance across several classes. ResNet50 shows significant inconsistencies, especially for Powdery Mildew and Sooty Mould, where it records lower precision and recall values (as low as 0.54 precision and 0.70 F1-score for Powdery Mildew). Similarly, EfficientNetB0 demonstrates relatively high scores overall but struggles with Bacterial Canker and Sooty Mould, where precision and recall drop to 0.84 and 0.81, respectively. AlexNet, VGG16, and VGG19 also achieve respectable performance but reveal slight variations, particularly for Sooty Mould and Gall Midge, where the F1-scores dip below 0.95. While VGG16 and VGG19 maintain decent consistency across classes, their performance still falls short compared to MangoLeafXNet. In summary, the proposed MangoLeafXNet achieves unparalleled consistency and superior classification capability across all categories, significantly outperforming traditional architectures like ResNet50 and EfficientNetB0, especially in handling challenging classes prone to misclassification.

Following the detailed performance metrics in Table 5, Table 6 presents a comparative analysis of various DL models, including AlexNet, VGG16, VGG19, ResNet50, EfficientNetB0, and MangoLeafXNet, evaluated on the MangoLeafBD dataset. Among these, MangoLeafXNet demonstrates the highest overall performance, achieving an evaluation score exceeding 99.5%. This result highlights its superior robustness compared to other models. While AlexNet and VGG16 maintain consistent performance, both achieving

TABLE 5. Performance comparison of different deep learning models in predicting mango leaf diseases on the MangoLeafBD dataset.

Models	Predicted Labels	Precision	Recall	F1-Score
AlexNet	Powdery Mildew	0.98	1.00	0.99
	Gall Midge	0.96	0.98	0.97
	Die Back	0.98	1.00	0.99
	Sooty Mould	0.90	0.96	0.93
	Anthraco nose	1.00	0.96	0.98
	Healthy	0.97	1.00	0.98
	Cutting Weevil	1.00	1.00	1.00
	Bacterial Canker	1.00	0.91	0.95
VGG 16	Powdery Mildew	0.96	0.96	0.96
	Gall Midge	0.93	1.00	0.96
	Die Back	1.00	1.00	1.00
	Sooty Mould	0.91	0.89	0.90
	Anthraco nose	1.00	0.95	0.97
	Healthy	1.00	1.00	1.00
	Cutting Weevil	1.00	1.00	1.00
	Bacterial Canker	0.96	0.96	0.96
VGG 19	Powdery Mildew	0.96	0.96	0.96
	Gall Midge	0.96	1.00	0.98
	Die Back	1.00	1.00	1.00
	Sooty Mould	0.85	0.96	0.90
	Anthraco nose	1.00	0.95	0.97
	Healthy	1.00	1.00	1.00
	Cutting Weevil	0.98	1.00	0.99
	Bacterial Canker	0.95	0.96	0.96
ResNet50	Powdery Mildew	0.54	1.00	0.70
	Gall Midge	0.87	0.87	0.87
	Die Back	1.00	1.00	1.00
	Sooty Mould	0.96	0.54	0.69
	Anthraco nose	1.00	0.95	0.97
	Healthy	1.00	1.00	1.00
	Cutting Weevil	1.00	1.00	1.00
	Bacterial Canker	1.00	0.89	0.94
EfficientNetB0	Powdery Mildew	0.92	0.94	0.93
	Gall Midge	0.91	0.98	0.94
	Die Back	1.00	0.96	0.98
	Sooty Mould	0.81	0.96	0.88
	Anthraco nose	0.98	0.98	0.98
	Healthy	0.95	1.00	0.97
	Cutting Weevil	0.98	0.91	0.94
	Bacterial Canker	1.00	0.84	0.91
MangoLeafXNet	Powdery Mildew	0.98	1.00	0.99
	Gall Midge	0.96	1.00	0.98
	Die Back	1.00	1.00	1.00
	Sooty Mould	1.00	0.96	0.98
	Anthraco nose	1.00	0.98	0.99
	Healthy	1.00	1.00	1.00
	Cutting Weevil	1.00	1.00	1.00
	Bacterial Canker	1.00	1.00	1.00

TABLE 6. Performance of the proposed MangoLeafXNet model against various deep learning models on the MangoLeafBD dataset.

Models	Accuracy	Precision	Recall	F1-score
AlexNet	98	98	98.25	98
VGG 16	98.33	97	97	97
VGG 19	97.23	96.25	96.12	96.12
ResNet50	88.33	91.12	85.37	86
EfficientNetB0	94.66	94.37	94.62	94.12
MangoLeafXNet	99.8	99.5	99.62	99.56

accuracy values close to 98%, ResNet50 and EfficientNetB0 display comparatively lower metrics. Specifically, ResNet50 records the lowest accuracy of 88.33%, coupled with a recall of 85.37% and F1-score of 86%, highlighting its reduced capability in handling the complexities of the MangoLeafBD dataset. Similarly, EfficientNetB0, though outperforming ResNet50, still lags behind MangoLeafXNet, achieving an accuracy of 94.66% and an F1-score of 94.12%.

TABLE 7. Classification performance of the proposed model for the Mango Pest Dataset.

Model	Predicted Labels	Precision	Recall	F1-Score
MangoLeafXNet	Ischnaspis Longirostris	1.00	1.00	1.00
	Ceroplastes Rubens	0.83	1.00	0.91
	Aulacaspis Tubercularis	1.00	1.00	1.00
	Erosomyia Sp	1.00	1.00	1.00
	Cisaberoptus Kenyae	1.00	1.00	1.00
	Orthaga Euadrusalis	1.00	1.00	1.00
	Procontarinia Rubus	0.90	1.00	0.95
	Icerya Seychellarum	1.00	0.82	0.90
	Procontarinia Matteiana	1.00	1.00	1.00
	Dialeuropora Decempuncta	1.00	1.00	1.00
	Neomelicharia Sparsa	1.00	1.00	1.00
	Normal	1.00	1.00	1.00
	Dappula Tertia	0.88	1.00	0.93
	Valanga Nigricornis	0.83	0.83	0.83
	Apoderus Javanicus	1.00	0.88	0.93
	Mictis Longicornis	1.00	0.88	0.93

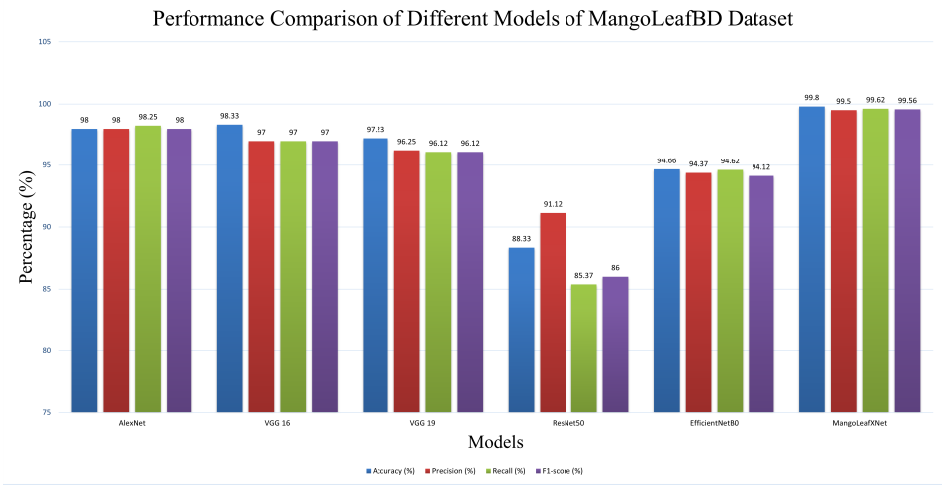


FIGURE 19. Performance Comparison of Different Models on Various Metrics.

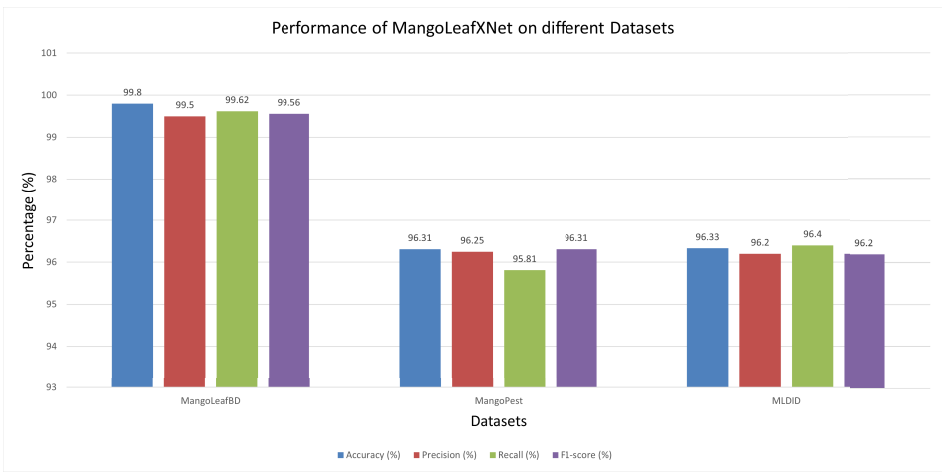


FIGURE 20. Performance of proposed MangoLeafXNet on different datasets.

This comprehensive analysis validates MangoLeafXNet’s effectiveness in accurately classifying mango leaf diseases.

Table 7 presents a performance evaluation of the MangoLeafXNet model on the Mango Pest Dataset, highlighting

TABLE 8. Classification performance of proposed model for the Mango Leaf Disease Identification Dataset.

Model	Predicted Labels	Precision	Recall	F1-Score
MangoLeafXNet	Gall Mid	0.95	0.92	0.94
	Die Back	0.93	0.98	0.95
	Anthrachnose	0.98	0.94	0.96
	Healthy	0.98	0.98	0.98
	Bacterial Canker	0.97	1.00	0.98

its classification effectiveness across different pest species using key evaluation metrics. The results highlight the model's strong performance, with several classes achieving perfect scores. For instance, *Aulacaspis Tubercularis*, *Cisaberoptus Kenyae*, and *Dialeuropora Decempuncta* all have perfect precision, recall, and F1-scores of 1.00. Other species, such as *Apoderus Javanicus* and *Icerya Seychellarum*, show strong performance with F1-scores above 0.90, though some show slightly lower recall values, such as *Icerya Seychellarum* with a recall of 0.82. Overall, these results underscore MangoLeafXNet's effectiveness in accurately classifying mango pests across a diverse set of categories. In a similar context, Table 8 offers a detailed look at MangoLeafXNet's performance on the Mango Leaf Disease Identification Dataset, further demonstrating its robust and consistent classification abilities across different datasets. The model achieves high precision and recall across various disease categories, with Bacterial Canker standing out for its perfect recall (1.00) and strong F1-score of 0.98. Diseases such as Anthracnose and Die Back also show impressive performance, with F1-scores of 0.96 and 0.95, respectively. The model performs equally well in classifying healthy leaves, achieving an F1-score of 0.98. These results reflect the model's versatility in effectively identifying both diseased and healthy mango leaves, solidifying its role as a reliable tool for mango leaf disease diagnosis.

Table 9 showcases the evaluation metrics of the proposed MangoLeafXNet model across three datasets: MangoLeafBD, MangoPest, and MLDID. The model delivers remarkable results on the MangoLeafBD dataset, achieving an overall evaluation score exceeding 99.5%, highlighting its effectiveness in classifying mango leaf diseases. On the MangoPest dataset, MangoLeafXNet continues to perform strongly, attaining an accuracy of 96.31%, along with precision, recall, and F1-score values of 96.25%, 95.81%, and 96.31%, respectively. Similarly, the model demonstrates consistent performance on the MLDID dataset, with an overall score surpassing 96% across all assessment metrics. These findings reinforce the model's adaptability and reliability, confirming its capability to generalize effectively across multiple datasets and real-world scenarios.

Furthermore, Figure 19 presents a comparative analysis of various DL models, including AlexNet, VGG16, VGG19, ResNet50, EfficientNetB0, and the proposed MangoLeafXNet, evaluated exclusively on the MangoLeafBD dataset. The results highlight the performance of MangoLeafXNet, achieving the highest evaluation scores among

all models, showcasing its effectiveness in accurately classifying mango leaf diseases. In addition, Figure 20 highlights the adaptability and generalizability of MangoLeafXNet across three datasets: MangoLeafBD, MangoPest, and Mango Leaf Disease Identification Dataset (MLDID). The model consistently delivers high evaluation scores across all datasets, with its peak performance observed on the MangoLeafBD dataset. Together, these figures validate MangoLeafXNet's robustness, reliability, and applicability in real-world agricultural scenarios, establishing it as a benchmark for automated mango leaf disease classification and management.

Table 10 outlines the classification performance of multiple models for identifying mango leaf diseases. Rizvee et al. [15] proposed the LeafNet model, achieving an accuracy of 99.55%, precision of 99.5%, recall of 99.45%, and an F1-score of 99.47%. However, this model does not incorporate XAI. Mahbub et al. [28] utilized the LCNN model, achieving an accuracy of 98%, along with precision, recall, and F1-scores of 97.6%, 97.5%, and 97.5%, respectively, also without the integration of XAI. Singh et al. [29] introduced the MCNN model, which achieved an accuracy of 97.13%. However, other evaluation metrics were not reported, and XAI was not employed. Vijay et al. [30] combined EfficientNetB4 with a Custom CNN architecture, achieving an accuracy of 93.01%, precision of 93.23%, recall of 94%, and an F1-score of 93.07%, similarly without incorporating XAI.

In contrast, the proposed MangoLeafXNet presented in this paper exhibits superior performance across multiple datasets. On the MangoLeafBD dataset, it achieves an accuracy of 99.8%, precision of 99.5%, recall of 99.62%, and an F1-score of 99.56%, surpassing all other models. Additionally, MangoLeafXNet demonstrates consistent and reliable performance on the MangoPest and MLDID datasets, achieving accuracies of 96.31% and 96.33%, respectively, with comparable precision, recall, and F1-scores. Unlike other models, MangoLeafXNet uniquely integrates XAI, enhancing interpretability by providing transparency and insights into its decision-making process.

In summary, the proposed MangoLeafXNet model sets a new benchmark for classifying mango leaf diseases, outperforming state-of-the-art models across multiple datasets and evaluation metrics. Its ability to generalize across different datasets and incorporate XAI for interpretability makes it a robust and practical solution for real-world agricultural applications. The presented results demonstrate the model's exceptional accuracy, reliability, and versatility, offering significant advancements in automated disease detection and management.

B. QUALITATIVE ASSESSMENT

1) GRAD-CAM

Gradient-weighted Class Activation Mapping (GRAD-CAM) is an essential technique in DL for visualizing the

TABLE 9. Performance of the proposed MangoLeafXNet model across different datasets.

Datasets	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
MangoLeafBD [25]	99.8	99.5	99.62	99.56
MangoPest [26]	96.31	96.25	95.81	96.31
MLDID [27]	96.33	96.2	96.4	96.2

TABLE 10. State-of-the-art comparison of models for classifying mango leaf diseases across multiple datasets, highlighting evaluation metrics, and XAI features.

Ref.	Model	Accuracy (%)	Precision	Recall	F1-score	XAI
Rizvee et al. [15]	LeafNet	99.55	99.5	99.45	99.47	✗
Mahbub et al. [28]	LCNN	98	97.6	97.5	97.5	✗
Singh et al. [29]	MCNN	97.13	-	-	-	✗
Vijay et al. [30]	EfficientNetB4 + Custom CNN	93.01	93.23	94	93.07	✗
This paper	MangoLeafXNet with MangoLeafBD	99.8	99.5	99.62	99.56	✓
	MangoLeafXNet with MangoPest	96.31	96.25	95.81	96.31	✓
	MangoLeafXNet with MLDID	96.33	96.2	96.4	96.2	✓

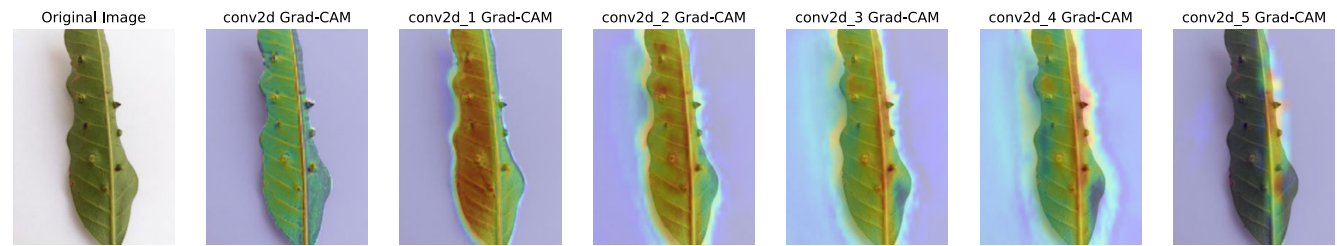


FIGURE 21. Grad-CAM output of Gall Midge for different convolutional layers of the proposed MangoLeafXNet model.



FIGURE 22. Grad-CAM output of Sooty Mould for different convolutional layers of the proposed MangoLeafXNet model.



FIGURE 23. Grad-CAM output of Die Back for different convolutional layers of the proposed MangoLeafXNet model.

regions of an image that significantly contribute to the model’s predictions. By generating heatmaps, GRAD-CAM highlights the areas the model focuses on to make decisions, enhancing interpretability. In this study, the proposed MangoLeafXNet model effectively utilizes GRAD-CAM to identify diseased regions on mango leaves or confirm their

absence, offering insightful information about how the model makes decisions.

Figure 21 illustrates the Grad-CAM outputs for Gall Midge across various convolutional layers. The visualizations reveal how the model’s focus evolves from broader areas in the earlier layers to more specific patterns in the deeper

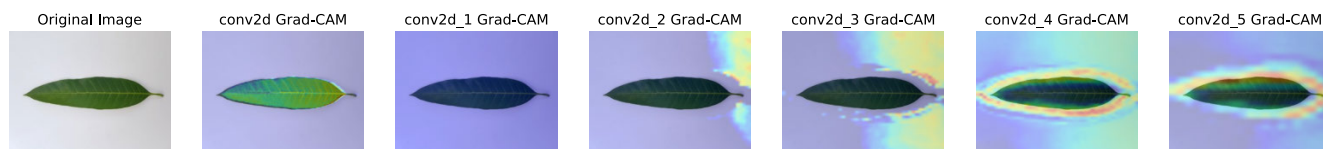


FIGURE 24. Grad-CAM output of Healthy leaf for different convolutional layers of the proposed MangoLeafXNet model.



FIGURE 25. Grad-CAM output of Neomelicharia Sparsa for different convolutional layers of the proposed MangoLeafXNet model.

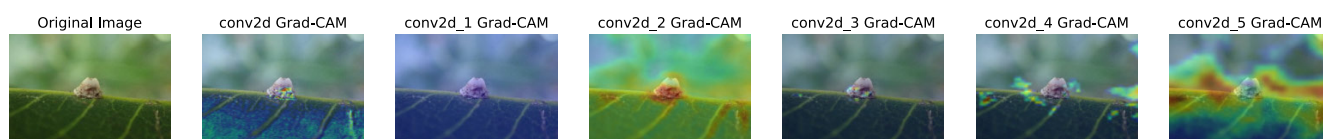


FIGURE 26. Grad-CAM output of Ceroplastes Rubens for different convolutional layers of the proposed MangoLeafXNet model.



FIGURE 27. Grad-CAM output of Normal leaf for different convolutional layers of the proposed MangoLeafXNet model.



FIGURE 28. Grad-CAM output of Bacterial Canker for different convolutional layers of the proposed MangoLeafXNet model.



FIGURE 29. Grad-CAM output of Die Back for different convolutional layers of the proposed MangoLeafXNet model.

layers, effectively capturing the distinctive features of this disease. Similarly, Figure 22 depicts Grad-CAM results for Sooty Mould, showcasing the model's ability to localize the affected regions as it processes the image through successive

layers. This progression demonstrates the refinement in the network's focus, improving classification accuracy for Sooty Mould. In Figure 23, the Grad-CAM visualization for Die Back highlights the characteristic features of the disease, with

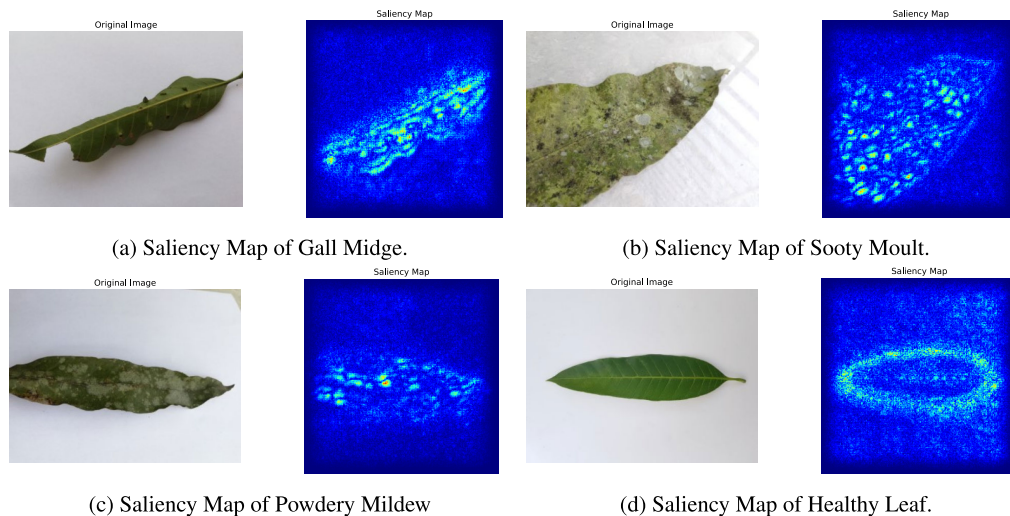


FIGURE 30. Saliency Maps highlighting key regions influencing MangoLeafXNet's classification for various mango leaf conditions.

deeper layers accurately identifying the affected areas on the leaf. In contrast, Figure 24 represents the Grad-CAM outputs for Healthy leaves, where the model's attention is distributed across the entire leaf surface. This uniform attention pattern indicates the absence of anomalies and reinforces the classification of the leaf as healthy. Further analysis using Grad-CAM is presented in Figure 25 for *Neomelicharia Sparsa*, where the model captures subtle textural patterns associated with this condition. The progression from broader attention in initial layers to more localized focus in deeper layers underscores the model's capacity for precise feature extraction. Similarly, Figure 26 visualizes Grad-CAM outputs for *Ceroplastes Rubens*, demonstrating how the network isolates clusters of affected regions for accurate classification. Figure 27 explores Grad-CAM results for Normal leaves, where the attention is uniformly distributed, reflecting the absence of significant disease markers and confirming the classification. Figure 28 highlights Grad-CAM for Bacterial Canker, showing a precise focus on the diseased regions with exceptional clarity in deeper layers. Finally, Figure 29 revisits the Grad-CAM outputs for Die Back, reaffirming the model's robustness and consistency in identifying features critical to this disease.

These Grad-CAM visualizations significantly contribute to enhancing the reliability of the model by providing transparent, visual explanations of its predictions. Instead of relying solely on numerical outputs, users can visually confirm whether the model's decision is based on relevant, disease-affected regions of the leaf. This interpretability allows agricultural researchers to validate the model's reasoning and ensures that the predictions align with domain knowledge about disease symptoms.

For end users such as farmers, the Grad-CAM heatmaps offer an intuitive way to understand why a leaf has been classified as diseased or healthy. By visually highlighting the specific areas of concern, farmers can better trust the model's diagnosis and take informed actions, such as

targeted treatment or isolation of affected plants. This visual feedback reduces the risk of misclassification, builds user confidence, and aids in real-world decision-making, bridging the gap between complex AI models and practical agricultural applications.

Therefore, integrating Grad-CAM not only supports the model's interpretability but also acts as a critical tool for increasing the transparency, reliability, and practical usability of the MangoLeafXNet system in agricultural settings.

2) SALIENCY MAP

Saliency maps are an effective visualization tool used to interpret DL models by highlighting the most influential pixels contributing to a prediction. In this study, saliency maps are employed to further enhance the interpretability of the proposed MangoLeafXNet model, providing a complementary perspective alongside Grad-CAM visualizations.

Figure 30 presents the saliency maps for various mango leaf conditions, including Gall Midge, Sooty Mould, Powdery Mildew, and Healthy leaves. Each subfigure pairs the original image with its corresponding saliency map, highlighting the regions that most influence the model's classification. For Gall Midge (30a), the saliency map emphasizes the central distorted areas of the leaf, matching the visible deformation patterns. In Sooty Mould (30b), the model focuses on scattered bright regions that align with the fungal patches seen on the leaf surface. Similarly, the Powdery Mildew saliency map (30c) clearly highlights areas with dense, powdery spots, indicating accurate attention to disease-affected regions. In contrast, the Healthy leaf saliency map (30d) shows an even, diffuse focus without any concentrated areas, confirming the absence of disease symptoms. These visualizations demonstrate how the model's attention aligns with actual disease markers, enhancing both interpretability and trust in the classification results.

The inclusion of saliency maps enhances the reliability of the model by offering transparent insights into its

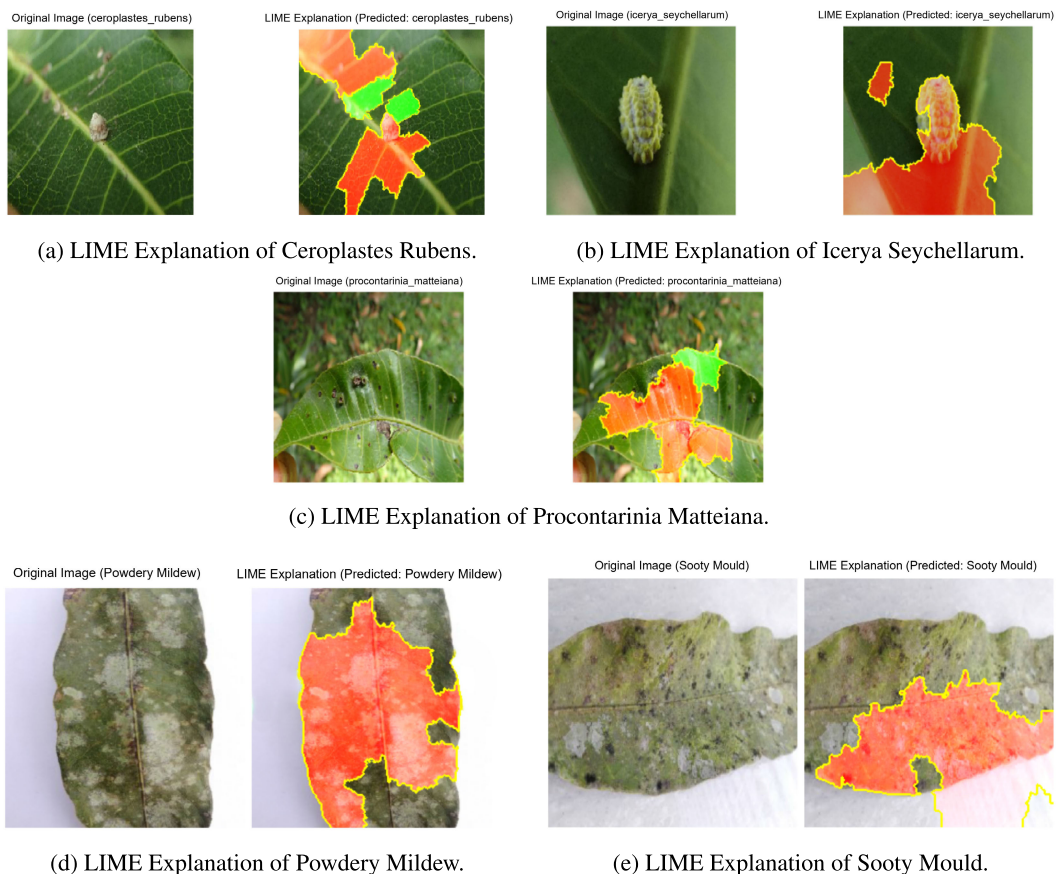


FIGURE 31. LIME explanations for different pest and disease predictions. Each subfigure shows the original image (left) and its corresponding LIME overlay (right), visualizing the model's local reasoning.

decision-making process. This visualization allows end users, such as farmers and agricultural researchers, to verify whether the model's attention aligns with known disease characteristics. By providing visual confirmation, saliency maps build user trust, facilitate practical diagnosis, and support informed agricultural decision-making.

3) LIME

While saliency maps are already used in our study to highlight global attention regions, we further employ LIME (Local Interpretable Model-agnostic Explanations) to gain more localized, instance-specific insights into the decision-making behavior of our model. LIME works by perturbing the input image and observing the model's response, allowing it to approximate which regions most strongly influence a specific prediction. Unlike saliency maps, LIME explanations are interpretable at a superpixel level and provide class-specific confidence shifts, helping to validate whether the model is responding to biologically relevant cues or misleading visual artifacts. Figure 31 illustrates a series of LIME interpretations across different pest and disease classes.

In Figure 31a, the input image features a visible *Ceroplastes rubens* pest concentrated near the central vein of the leaf. The corresponding LIME overlay highlights a broad region

around the pest, extending into the surrounding healthy tissue. While the model does partially detect the actual pest, the highlighted region suggests it is relying on contextual cues like shadows and discoloration rather than precisely isolating the insect. This demonstrates that although the model is responsive to the anomaly, its localization remains coarse. Figure 31b presents an image of a leaf with *Icerya seychellarum*. The model again succeeds in attending to the central area where the pest is located, but the LIME mask extends beyond the pest body and marks large areas of the surrounding leaf surface. This indicates a misinterpretation by our proposed model. In Figure 31c, the image illustrates subtle leaf deformation attributed to *Procontarinia matteiana*. The LIME output highlights a wide region across the leaf, including some areas without overt visual symptoms. This behavior is reflecting the model's sensitivity to distributed patterns of discoloration and minor deformations, even when they are not sharply delineated. The LIME output for powdery mildew, shown in Figure 31d, demonstrates a focused correspondence between the white-gray fungal patches and the high-importance areas identified in the explanation. The overlay aligns well with visibly infected zones, suggesting a strong association between these visual patterns and the model's internal representation of powdery

mildew. Figure 31e corresponds to an image affected by sooty mould. Prominent dark fungal growth is present, and the LIME explanation covers much of this region. Some attention also extends into surrounding leaf areas, which may result from overlapping texture gradients or shared visual context across adjacent foliage.

While other XAI techniques such as Grad-CAM and saliency maps demonstrate consistent and well-localized attention on our model, the LIME-based explanations exhibit certain limitations. Although LIME successfully identifies the general regions associated with pests and diseases, the highlighted areas often extend beyond the visibly affected zones. This tendency to overextend may stem from the superpixel-based approximation process and its sensitivity to texture continuity or surrounding context. As a result, while LIME provides valuable interpretability at the local level, its spatial precision in delineating exact symptomatic boundaries is comparatively lower, warranting careful interpretation in fine-grained diagnostic tasks.

4) GRADIO WEB DEPLOYMENT

To evaluate the performance of the proposed MangoLeafXNet model, we developed an interactive interface using Gradio, an open-source Python library designed for real-time classification and model validation. This web-based interface offers a user-friendly experience with a drag-and-drop or file-upload option, allowing users to simply upload an image of a mango leaf or pest and receive instant feedback.

Once an image is uploaded, it is automatically resized to 227×227 pixels, preprocessed, and passed through the trained model. The interface then displays the predicted class label along with a confidence score, which is visually overlaid on the image using the Pillow (PIL) library for intuitive interpretation. This enables users—especially those without technical expertise—to understand the diagnosis at a glance.

The interface streamlines the entire classification process, from input to visualization, making it practical for real-time usage. Figures 32a and 32b illustrate successful classifications of mango pest and healthy leaves. The model accurately identified *Icerya seychellarum* with 99.98% confidence and a healthy normal mango leaf with 99.97% confidence. Additionally, Figures 32c and 32d demonstrate the model's capability to classify Anthracnose disease and Cutting Weevil infestation with 99.99% and 100.00% confidence, respectively.

Currently, this interface runs locally on a localhost server, making it accessible only from the host machine. While this serves as a valuable tool for internal validation and testing, broader accessibility is required for real-world adoption. To address this, we plan to deploy the interface on a publicly accessible platform such as Hugging Face Spaces, Google Cloud, or AWS. This will allow researchers, farmers, and agricultural experts to interact with the model remotely, eliminating the need for local installations and improving usability.

Future improvements include adding support for batch processing, integrating Grad-CAM heatmaps for better model interpretability, and optimizing inference time for mobile applications. By deploying this system on the cloud, we aim to make MangoLeafXNet a widely accessible and scalable tool for real-world agricultural disease detection.

5) INDEPENDENT VALIDATION

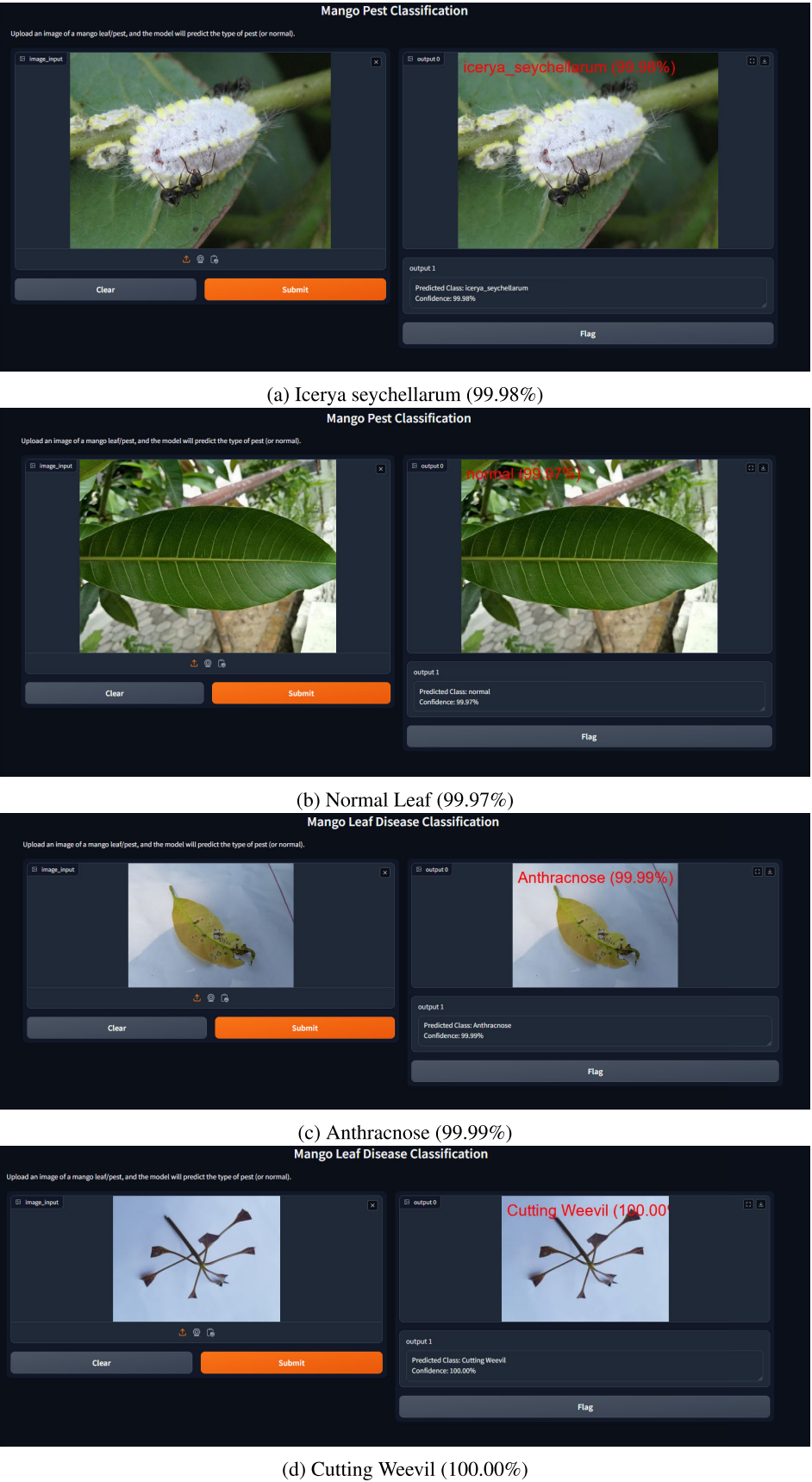
Inspired by the work of Bhandari et al. [31], who conducted independent validation using real-time tomato leaf images from external datasets, we also performed an independent analysis to further evaluate the generalization capability of the proposed MangoLeafXNet model. Independent analysis refers to testing the model on images that were never seen during training, validation, or testing phases, ensuring that the model performs well on entirely new, real-world data.

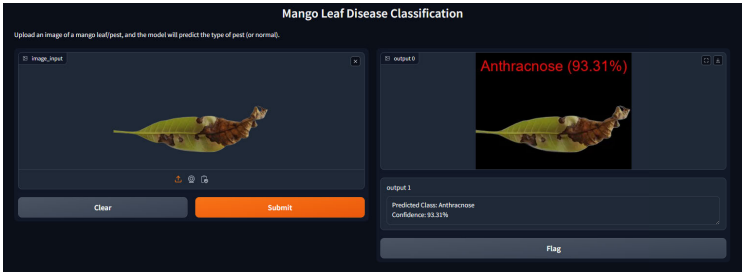
For this purpose, we collected a set of random mango leaf images from the publicly available Mango Leaf Dataset [32]. A set of random images representing different disease classes and healthy leaves were selected from this independent dataset, including Anthracnose, Gall Midge, Powdery Mildew, Sooty Mould, and Healthy samples. These images were resized appropriately and processed through our Gradio-based real-time MangoLeafXNet deployment.

The model accurately predicted the respective classes with high confidence levels, demonstrating its robustness when exposed to unseen data. Specifically, Anthracnose was detected with a confidence of 93.31% (33a), Gall Midge with 85.91% (33b), Healthy leaf with 89.76% (33c), Powdery Mildew with 99.97% (33d), and Sooty Mould with 93.80% (33e). These results validate the generalization ability of the proposed model beyond controlled dataset conditions. Representative classification results for the independently collected images are illustrated in Figure 33.

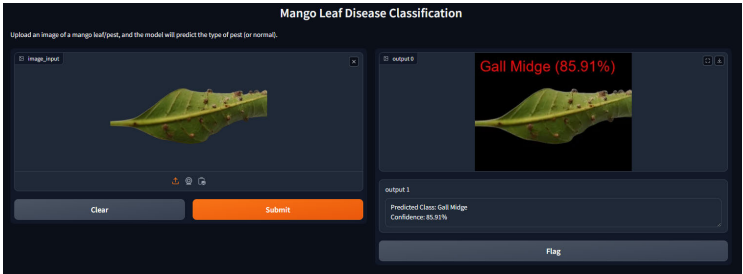
V. DISCUSSION AND LIMITATIONS

DL algorithms for image analysis in agriculture have become crucial for detecting crop diseases. These diseases, being major contributors to crop destruction, have significant economic impacts worldwide. DL algorithms provide quick and precise identification, enabling timely intervention and mitigation. This study presents the MangoLeafXNet architecture, specifically designed for classifying mango leaf diseases. The model was evaluated on the MangoLeafBD dataset, achieving exceptional classification performance with an accuracy of 99.8%, precision of 99.5%, recall of 99.62%, and an F1-score of 99.56%, significantly outperforming established models like AlexNet, VGG16, and VGG19 under the same hyperparameter settings. To further validate the robustness and generalizability of MangoLeafXNet, the model was tested on two additional datasets: the MangoPest dataset and the Mango Leaf Disease Identification Dataset (MLDID). Across these datasets, the model maintained consistently high performance, demonstrating its adaptability and reliability in diverse agricultural contexts. These results

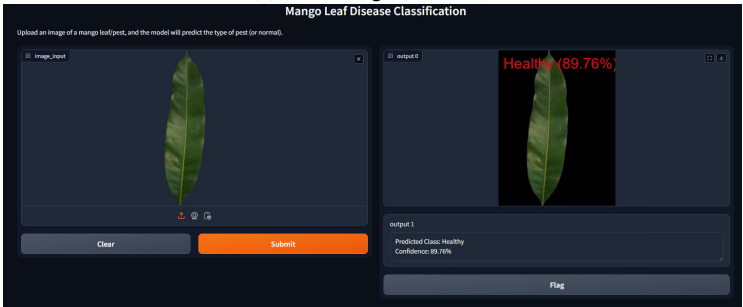




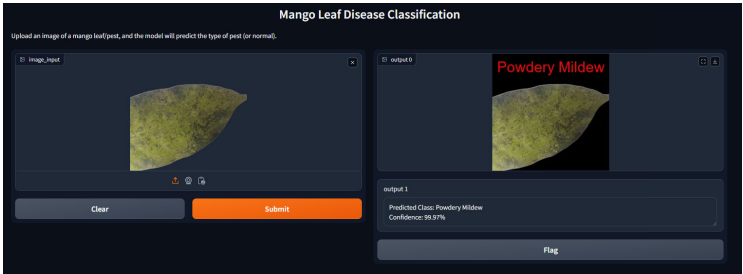
(a) Anthracnose (93.31%).



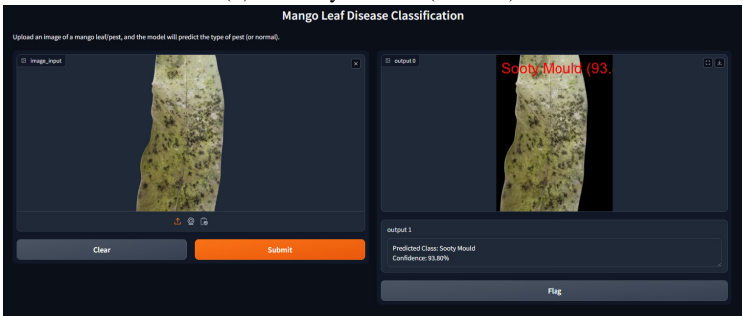
(b) Gall Midge (85.91%).



(c) Healthy (89.76%).



(d) Powdery Mildew (99.97%).



(e) Sooty Mould (93.80%).

FIGURE 33. Independent validation of the MangoLeafXNet model on unseen mango leaf images from an external dataset.

highlight MangoLeafXNet's potential as a valuable tool for real-world agricultural disease management and precision farming applications.

One of the key strengths of the MangoLeafXNet architecture is its ability to capture intricate disease patterns, making it particularly effective for the domain-specific challenge of mango leaf disease classification. This performance is achieved through an optimized combination of convolutional layers, batch normalization, and dropout regularization, which together enhance the model's ability to generalize from the training data. The superior accuracy of MangoLeafXNet underscores the potential of tailored CNN architectures for specific agricultural applications, where domain-specific features are critical for accurate classification. The qualitative assessments using GRAD-CAM, saliency map, and LIME provide insights into the interpretability of the model. These visualization techniques reveal the specific regions of the leaf images that the model focuses on when making predictions, ensuring transparency in the decision-making process. This interpretability is crucial for gaining the trust of end-users, such as farmers and agricultural experts, who rely on these systems for disease diagnosis. To facilitate real-time classification and model validation, we developed an interactive Gradio-based web interface. This interface enables users to upload mango leaf images and receive instant classification results with confidence scores, enhancing accessibility for researchers and agricultural professionals. Currently, the system runs locally on a localhost server, allowing for internal testing and validation.

Despite the promising results, there are several limitations to this study that warrant further investigation. The proposed MangoLeafXNet model focuses solely on image-based classification. In real-world scenarios, additional factors such as soil health, weather conditions, and pest presence may influence disease occurrence and severity. Integrating multimodal data, including environmental and sensor data, could improve the accuracy and reliability of disease diagnosis systems. Another limitation is the computational complexity of the proposed model. MangoLeafXNet requires significant computational resources for training, which may limit its accessibility for users with limited hardware capabilities. Exploring ways to optimize the model or developing lighter versions without compromising much on accuracy could enhance its practical usability in resource-constrained environments. A further limitation is observed in the LIME explanations. While they correctly highlight relevant regions, the highlighted areas often extend beyond the actual diseased parts, indicating a lack of spatial precision in certain cases.

Future research can focus on several key areas to further enhance the effectiveness and applicability of the MangoLeafXNet architecture. One promising direction is the integration of multimodal data, combining image-based analysis with other relevant data sources such as environmental conditions, soil health, and pest activity. This holistic approach will improve the model's accuracy and reliability in real-world agricultural settings. Additionally, efforts will

be made to optimize the model's architecture to reduce its computational demands, making it more accessible for deployment in resource-constrained environments. Transfer learning and ensemble methods will also be explored to boost the model's robustness and generalizability across different regions and disease variants. Furthermore, expanding MangoLeafXNet beyond its current localhost deployment to a fully web-based application will be a key focus of future research. This transition will enable real-time disease diagnosis and broader accessibility for farmers, researchers, and agricultural professionals without requiring local installations. The deployment process will involve hosting the model on cloud platforms such as Hugging Face Spaces, Google Cloud, or AWS, ensuring scalability and seamless remote access. Additionally, efforts will be made to develop an intuitive and responsive user interface, optimized for both desktop and mobile devices, to enhance usability in real-world agricultural settings. Ensuring the system can handle large-scale, real-time data processing while maintaining high-speed inference and reliability will be crucial for practical deployment. By addressing these aspects, future research aims to bridge the gap between AI-driven disease classification and real-world agricultural applications, contributing to precision farming and sustainable crop health management.

VI. CONCLUSION

In conclusion, this study presents a CNN architecture designed specifically for the classification of mango leaf diseases, contributing to advancements in automated disease diagnosis for mango cultivation. Through rigorous experimentation and comparative analysis, the MangoLeafXNet model demonstrated exceptional performance across three datasets. On the primary MangoLeafBD dataset, the model achieved a remarkable accuracy of 99.8%. Further validation on the MangoPest dataset and the Mango Leaf Disease Identification Dataset (MLDID) achieved accuracies of 96.31% and 96.33%, respectively, confirming the model's robustness and adaptability in diverse agricultural contexts. The implications of these findings extend beyond academic research, offering promising applications for real-world mango cultivation and disease management. By enabling rapid and objective disease diagnosis, the automated classification system provides farmers with timely insights for targeted interventions, mitigating yield losses and improving crop health outcomes. Looking ahead, future research may focus on optimizing the model for deployment in resource-constrained environments by exploring pruning techniques or developing lightweight versions to reduce computational demands without significantly compromising accuracy. Additionally, the integration of state-of-the-art ML techniques, such as transfer learning and ensemble methods, could further enhance the robustness and generalizability of the model. Deploying MangoLeafXNet as a web-based application with real-time diagnostic capabilities also represents a promising direction for broadening its accessibility to farmers and agricultural professionals.

In summary, MangoLeafXNet presents a practical, high-performing solution for mango leaf disease classification, with strong potential to enhance disease management and promote sustainable mango cultivation.

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